Article

A novel bacterial protein family that catalyses nitrous oxide reduction

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Nitrous oxide (N_2O), a driver of global warming and climate change, has reached unprecedented concentrations in Earth's atmosphere[1]. Current N_2O sources outpace N_2O sinks, emphasizing the need for comprehensive understanding of processes that consume N_2O. Microbes that express the enzyme N_2O reductase (N_2OR) convert N_2O to climate change-neutral dinitrogen (N_2). Known N_2ORs belong to the canonical clade I and clade II NosZ reductases and are considered key enzymes for N_2O reduction[2–4]. Here we report a previously unrecognized protein family with a role in N_2O reduction, clade III lactonase-type N_2OR (L-N_2OR), which diverges in sequence from canonical NosZ but conserves three-dimensional protein structural features. Integrated physiological, metagenomic, proteomic and structural modelling studies demonstrate that L-N_2ORs catalyse N_2O reduction. L-N_2OR genes occur in several phyla, predominantly in uncultured taxa with broad geographic distribution. Our findings expand the known diversity of N_2ORs and implicate previously unrecognized taxa (for example, Nitrospinota) in N_2O consumption. The expansion of N_2OR diversity and the identification of a novel type of catalyst for N_2O reduction advances the understanding of N_2O sinks, has implications for greenhouse gas emission and climate change modelling, and expands opportunities for innovative biotechnologies aimed at curbing N_2O emissions[5,6].

N_2O is a greenhouse gas with substantially higher warming potential than carbon dioxide and methane. Beyond trapping heat in the atmosphere, N_2O exacerbates warming by depleting stratospheric ozone[7]. N_2O concentrations reached 332 parts per billion by volume in 2019, a level that is likely to be unprecedented in Earth's atmosphere[7,8], with an accelerating trend over the past 50 years. At the root of increasing N_2O emissions lies one of the most consequential inventions of the twentieth century: the Haber–Bosch chemical process, which enables the large-scale production of synthetic nitrogen fertilizers[9]. Global production of synthetic nitrogen fertilizers (estimated at 210 Tg yr^{-1}) is essential for provisioning the food needed to sustain Earth's human population and exceeds the amount of N_2 that is captured naturally in agroecosystems through biological N fixation[10] (203 Tg yr^{-1}). This observation implies that about half of the N atoms in a human living in an industrialized nation have passed through a chemical plant running the Haber–Bosch process, highlighting the major anthropogenic disturbance that N fertilizers exert on the N cycle[11]. Anthropogenic activities, particularly excessive application of N fertilizers coupled with limited N use efficiency of relevant crops (less than half of the N in fertilizer is taken up by the plant), have given rise to nitrate leaching and eutrophication, volatilization loss and conversion of reactive N to N_2O[12].

Microbial processes, in particular denitrification and nitrification[13,14], and coupled biotic–abiotic chemodenitrification[15] are major biogeochemical processes that generate N_2O. Microorganisms with N_2OR activity express canonical NosZ (C-NosZ), the catalyst that converts N_2O to environmentally benign N_2, often in a respiratory (that is, growth-linked) manner[4] (Box 1). At least some diazotrophs can utilize N_2O as an N source[16]; however, this process has a minor impact on the overall N_2O budget[17]. The continuous increase of atmospheric N_2O concentrations indicates an imbalance between global N_2O sources and sinks[18]. Optimizing N fertilizer management and improving N use efficiency can reduce N_2O emissions[19], and enhancing the natural N_2O sink capacity holds great promise for emission reductions[5,6]. This goal requires a comprehensive understanding of N_2OR types and the taxonomic diversity of N_2O-reducing microorganisms.

An unrecognized N_2OR

Efforts aimed at unravelling the microbiology of N_2O consumption in acidic tropical soils (Luquillo Experimental Forest, Puerto Rico) yielded anoxic Sabana soil microcosms with robust N_2O reduction activity maintained under low-N_2O (approximately 80 μmol N_2O consumed per year) and high-N_2O (around 3,000 μmol N_2O consumed per year) feeding regimens[20] at pH 4.5. Short-read metagenome sequencing recovered about 35-fold fewer C-*nosZ* sequences from the Sabana soil microcosm that consumed nearly 40-fold more N_2O under a high feeding regimen

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Box 1

Major biological N_2O sinks and terminology

Microorganisms with N_2OR activity reduce N_2O to N_2, mainly for the purpose of respiratory energy conservation[25], to potentially cope with N_2O inhibition of cob(I)amide-dependent enzymatic reactions[63,64], to dispose of excess electrons[65] or to facilitate plant infection[66]. Traditionally, N_2O reduction to N_2 was attributed to denitrifying microorganisms expressing NosZ, a concept that has been revised following the recognition of pathway modularity and the discovery of a distinct NosZ phylotype (clade II)[2,3]. Phylogenetic analyses differentiate clade I and clade II NosZ, with clade I often associated with complete denitrifiers. Here we define clade I and clade II *nos* gene clusters, *nosZ* genes and NosZ proteins as canonical. C-NosZ is encoded by the C-*nosZ* gene, and the catalytic function of the enzyme requires a set of similar accessory proteins encoded by the C-*nos* gene cluster.

C-NosZ enzymes contain two conserved copper-binding sites, Cu_A and Cu_Z, a distinguishing feature that sets C-NosZ apart from other reductases. The Cu_A site possesses a conserved CXXFCXXXHXEM motif, and Cu_Z has six conserved motifs, DXHH, XXHX, PHG, GPLH, XXGH and EPH.

compared with a replicate microcosm under a low-N_2O feeding regimen (Extended Data Fig. 1). This unexpected finding triggered efforts to derive a solid-free enrichment culture from the high-N_2O feeding regimen microcosm. Efforts to recover N_2O reduction activity in transfer cultures were successful following the replacement of lactate with 5 mM fumarate, and the addition of 4.16 mM (nominal; 416 μmol) hydrogen and tryptic soy broth (TSB) (20 mg l^{-1}) to the basal salt growth medium (Extended Data Fig. 2 and Supplementary Note 1). Following 25 monthly transfers (3%, v/v), the enrichment culture, designated Sab, consumed N_2O at similar rates of 93 ± 22.4 μmol per day per vessel (mean ± s.d. of 12 replicate batch incubations) at pH 4.5, 5.5 and 6.5, with no N_2O reduction observed at pH 7 or above (Supplementary Fig. 1 and Supplementary Note 1). Notably, no C-*nosZ* sequence reads were recovered from deeply sequenced metagenomic datasets of replicate Sab enrichment cultures grown with and without N_2O when the commonly used sequence cut-off values of 40% or 35% amino acid identity were applied to identify protein homologues[21,22] (Supplementary Information Note 2 and Supplementary Tables 1 and 2). When the sequence screening criterion was relaxed to 30% amino acid identity, a putative gene coding for N_2OR was found in the co-assembly (Fig. 1a and Supplementary Table 3).

Population genome binning revealed a single-type putative N_2OR gene assigned to a metagenome-assembled genome (MAG) representing a member of the genus *Desulfitobacterium* (Supplementary Note 3 and Supplementary Tables 4 and 5). Phylogenomic reconstruction of 31 *Desulfitobacteriaceae* genomes indicated that the MAG represents a novel *Desulfitobacterium* species, designated *Desulfitobacterium nosdiversum* strain Sab5 (Extended Data Fig. 3 and Supplementary Note 3). Light microscopy imaging illustrated the characteristic *Desulfitobacterium* cell morphology in cultures grown with N_2O, but not in cultures grown without N_2O (Supplementary Fig. 2). The majority (80.1%) of the metagenomic reads generated from the solid-free Sab enrichment cultures grown with N_2O could be matched to the strain Sab5 genome, whereas merely 1.74% of reads from Sab enrichment cultures grown with fumarate, TSB and H_2 but without N_2O matched the Sab5 genome (Fig. 1b). About 36% of the metagenomic reads of the initial Sabana soil microcosm matched the strain Sab5 genome, with no matching sequences detected in a replicate microcosm maintained without N_2O (Fig. 1c), revealing that strain Sab5 grew substantially during N_2O enrichment to dominate the microcosm and Sab enrichment culture microbial communities. Together, these results suggest that a novel *Desulfitobacterium* sp. carrying an unusual N_2OR-coding gene utilizes N_2O as electron acceptor at pH 4.5.

The putative N_2OR identified in *D. nosdiversum* strain Sab5 shared less than 35% amino acid identity with any protein sequences in the eggNOG v.5.0 database. A BLAST query against the NCBI non-redundant protein sequence database identified the top 100 protein sequences with an E-value of less than 10^{-5} for further refining the functional annotation of the putative N_2OR (Fig. 1a and Supplementary Table 6). On the basis of AlphaFold modelling, most of the retrieved protein sequences (n = 94) share structural similarity to lactonases with a characteristic 7-bladed β-propeller domain[23]. The amino acid identity between these lactonase-like proteins and the putative N_2OR of strain Sab5 ranges from 38 to 75%, and all possess a conserved Cu_A motif that is known to bind two Cu ions, a characteristic feature of C-NosZ[24]. Also present is a Cu_Z site that lacks the characteristic DXHH motif of C-NosZ[2,25]. On the basis of sequence differences (at most 35% amino acid identity to the C-NosZ sequences) and the lack of the characteristic DXHH motif in the Cu_Z site, here we refer to the putative N_2OR of *D. nosdiversum* strain Sab5 and the 94 lactonase-like proteins as lactonase-NosZ (L-NosZ) and refer to the gene as L-*nosZ*.

L-*nosZ* gene copy numbers increased in Sab enrichment cultures that received N_2O, but not in replicate cultures without N_2O (Fig. 1d). N_2O reduction did not occur in replicate cultures that received 6 ml acetylene (10%, v/v), a known inhibitor of C-NosZ activity[26]. Enumeration of the L-*nosZ* gene by quantitative PCR (qPCR) revealed a positive correlation between L-*nosZ* abundances and the amounts of N_2O consumed (Extended Data Fig. 4a), and we determined a growth yield of (6.86 ± 0.77) × 10^9 cells (4.21 ± 0.48 mg dry weight) per mmol N_2O consumed. This growth yield of Sab5 strain resembles yields reported for other soil-derived N_2O reducers, such as clade I *nosZ*-carrying *Bradyrhizobium* sp.[27] (8.3 × 10^9 cells (2.6 mg dry weight) per mmol N_2O), clade II *nosZ*-carrying *Anaeromyxobacter dehalogenans*[2] (4.08 × 10^8 cells (11.2 mg dry weight) per mmol N_2O), and *Desulfosporosinus nitrosoreducens*[25] (3.1 × 10^8 cells (0.19 mg dry weight) per mmol N_2O). Metaproteomic analysis of the N_2O-reducing Sab enrichment cultures revealed L-NosZ among the top ten most abundant proteins and detected a MacB-type ATP-binding cassette (ABC) transporter and *b*- and *c*-type cytochromes, which are encoded on genes adjacent to L-*nosZ* (Extended Data Fig. 4b).

The putative L-*nosZ* sequences (n = 94) retrieved from the NCBI database belong to uncultured bacteria, with one exception: the L-*nosZ* of the homoacetogenic bacterium *Sporomusa acidovorans* strain Mol (DSM3132[T])[28]. Reconstruction of the strain Mol genome resulted in a circular genome of 5.96 Mb, which does not contain C-*nosZ* and C-*nos* accessory genes, and this bacterium would therefore not be expected to reduce N_2O. In basal salt medium amended with yeast extract (1 g l^{-1}), strain Mol consumed two repeated N_2O feedings, after which N_2O reduction ceased. The addition of H_2 quickly restored N_2O consumption (Supplementary Fig. 3). N_2O consumption rates increased from 0.88 ± 0.03 to 1.73 ± 0.09 μmol N_2O h^{-1} per vessel during growth (Fig. 2a). Strain Mol did not grow in vessels with air headspace (21.3 kPa O_2, v/v) but growth was observed in vessels with 3,550 Pa O_2. N_2O reduction occurred at an O_2 partial pressure of 355 Pa but not of 709 Pa, indicating O_2 inhibition. Growth of strain Mol was observed between pH 4.5 to 7.5, but N_2O reduction occurred only between pH 6.5 to 7.5 (Extended Data Fig. 5a–d). Proteomic analysis comparing strain Mol grown with and without N_2O demonstrated that L-NosZ and eight proteins encoded on genes in proximity to the L-*nosZ* locus were exclusively detected in cells grown in the presence of N_2O (Fig. 2b). *Sporomusa ovata* DSM2662[T], a close relative of *S. acidovorans* strain Mol, lacks both C- and L-*nosZ* genes and was unable to reduce N_2O (Fig. 2c).

S. acidovorans strain Mol requires yeast extract for growth, complicating direct measurements of increases in cell number linked to N_2O

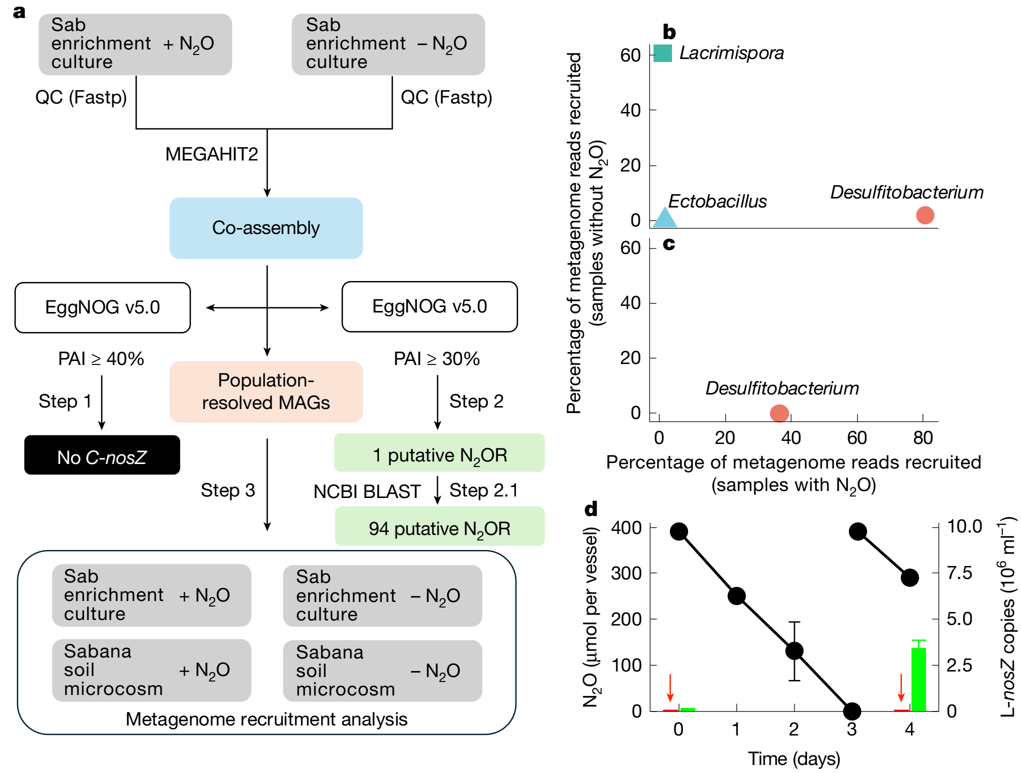

Fig. 1 | Identification of L-NosZ in Sab enrichment cultures. a, Workflow leading to the discovery of L-NosZ. Comparative analysis of metagenome datasets from Sab enrichment cultures maintained with N_2O (+N_2O) and without N_2O (−N_2O) identified the population responsible for N_2O reduction. Step 1: the eggNOG database v.5.0 was searched for C-NosZ with the commonly used percent amino acid sequence identity (PAI) cut-off values of 40% and 35%. Step 2: repeated searches against the eggNOG database v.5.0 with progressively relaxed percent amino acid identity cut-off values down to 30%. From Step 2, a putative gene encoding an N_2OR was identified in the co-assembly, which was compared against the NCBI non-redundant database to obtain closely related protein sequences (step 2.1). Step 3: recruitment analysis of metagenomes generated from Sab enrichment cultures and Sabana soil microcosms was performed with population-resolved MAGs as reference genomes. **b**,**c**, Percentage of metagenomic reads recruited to MAGs derived from the Sab enrichment cultures (**b**) and the Sabana soil microcosms (**c**). **d**, N_2O reduction performance of the Sab enrichment culture (N_2O, black dots) and L-*nosZ* gene copy numbers determined by qPCR (green bars). No increase in L-*nosZ* gene copies occurred in growth medium without N_2O (red bars indicated by the red arrows for day 0 and day 4). N_2O disappearance in autoclaved controls was not apparent. The increase of L-*nosZ* gene copies in Sab enrichment cultures strictly depended on the presence of N_2O in the cultivation vessels. Transfer cultures received a 3% (v/v) inoculum with $(8.17 \pm 1.9) \times 10^7$ *D. nosdiversum* cells. Data in **d** are mean ± s.d. of triplicate batch incubations. Where error bars are not visible, they are smaller than the symbols or borders.

respiration. In strain Mol cultures grown without N_2O, yeast extract fermentation and H_2/CO_2 reductive acetogenesis (with H_2 derived from fermentation reactions) are possible energy-conserving processes, and acetate production serves as a measure of growth under these incubation conditions[28]. Strain Mol cultures that had consumed different amounts of N_2O reached similar values for optical density at 600 nm, but produced substantially less acetate from yeast extract compared with cultures that did not receive N_2O (Extended Data Fig. 5e–g). A negative correlation between N_2O consumption and acetate production was observed (Extended Data Fig. 5h), suggesting that N_2O reduction diverts reducing equivalents from acetogenesis. Further studies using defined medium are needed to conclusively determine whether strain Mol utilizes N_2O as respiratory electron acceptor. In the presence of 6 ml acetylene (v/v; 10% of headspace), *S. acidovorans* grew but no N_2O was reduced. Growth and N_2O consumption did not occur in heat-killed *S. acidovorans* cultures, ruling out abiotic or physical processes leading to N_2O loss.

AlphaFold modelling of N_2ORs

C-NosZ and L-NosZ share less than 35% amino acid identity, making sequence-based functional predictions unreliable[21]. To reconcile low amino acid identity with shared function, we performed structural modelling of N_2ORs. Catalytically active C-NosZ features head-to-tail homodimers, each with a mixed-valent Cu_A site and a conserved tetranuclear Cu_Z site[24,29,30]. Previous crystallographic studies resolved the structures of clade I-type C-NosZ[24,29,30], and AlphaFold 2 (ref. 31) and AlphaFold 3 (ref. 32) were used here to investigate structural and functional features of L-NosZ. The structural models reveal characteristic features of clade I C-NosZ (Supplementary Fig. 4) and have predicted template modelling (pTM) scores and interface predicted template modelling (ipTM) scores exceeding 0.85, strongly supporting a correct backbone fold and oligomerization state. Superposition of L-NosZ models onto a C-NosZ crystal structure (*Pseudomonas stutzeri*, Protein Data Bank (PDB) entry 3SBQ[24]) revealed conserved amino acid residues in the Cu_A and Cu_Z sites. The two Cu_A sites in homodimeric crystal structure 3SBQ each comprise two histidine residues, two cysteine residues, and a methionine (Extended Data Fig. 6a), and these conserved amino acids are also present in the models of L-NosZ in similar coordinating positions (Extended Data Fig. 6b,c). Crystallographic studies have shown that the Cu_Z site is coordinated by seven histidine residues[24,29], which are also found in the L-NosZ models (Extended Data Fig. 6d–f). Root mean square deviations (RMSDs) of 0.6 to 2.0 Å were obtained for six paired histidine residues. Of note, H126 in the *S. acidovorans* L-NosZ and H130 in the *P. stutzeri* C-NosZ originate from different β-strands. Although their positions and relative orientations differ, these residues are expected to coordinate Cu in a similar manner and preserve the overall structure and function of the Cu_Z site. Together, the structural modelling, physiological, metagenomic, proteomic and metaproteomic investigations indicate that L-NosZ is a catalyst for N_2O reduction.

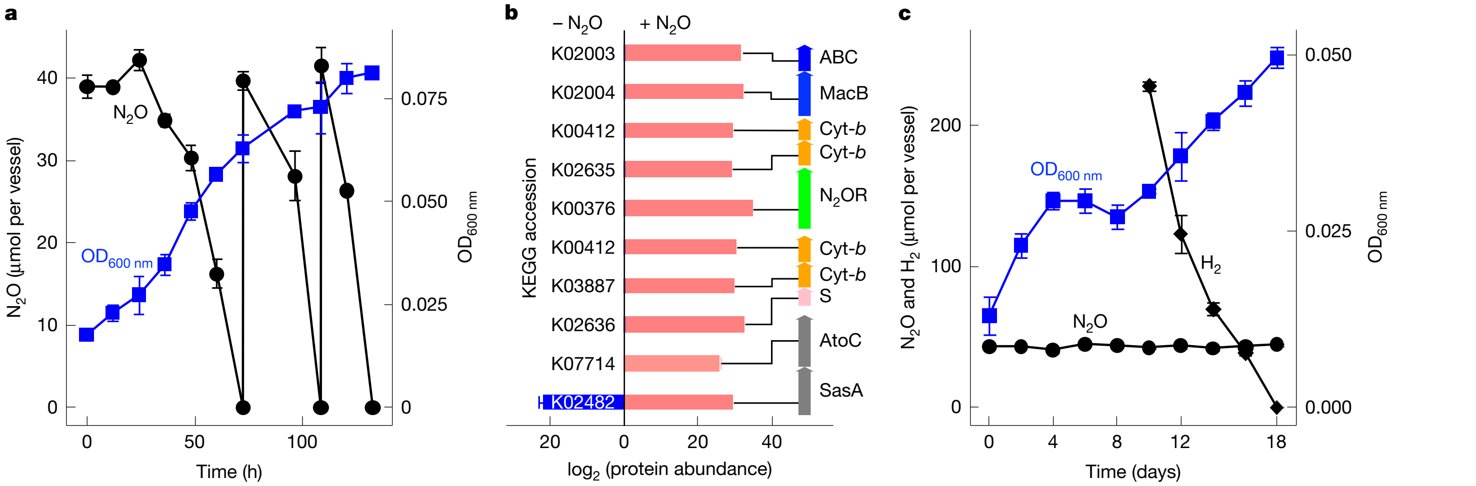

Fig. 2 | *S. acidovorans* strain Mol expresses L-NosZ to reduce N_2O. a, Growth (optical density at 600 nm ($OD_{600\,nm}$)) and accelerating N_2O reduction rates in *S. acidovorans* strain Mol cultures following repeated additions of N_2O in basal salt medium containing yeast extract. **b**, Abundance of L-NosZ and nine proteins encoded by genes neighbouring L-*nosZ* in *S. acidovorans* strain Mol cultures grown in medium containing yeast extract with and without N_2O. L-NosZ and eight out of the nine proteins were exclusively identified (pink bars) in cultures that received N_2O. The detection limit of this mass spectrometry measurement was approximately $\log_2$(protein abundance) = 14, and 8 out of the 9 proteins in strain Mol cultures with only yeast extract (no N_2O) were not detected above this threshold. Among the proteins of interest, only SasA (KEGG orthologue (KO) ID 2482, blue bar) was detected in strain Mol cultures grown without N_2O. The coloured vertical arrows display the arrangement of L-*nosZ* and neighbouring genes in the genome of *S. acidovorans* strain Mol. KO IDs of L-NosZ and proteins were inferred by comparison against the eggNOG database v.5 with the Cluster of Orthologous Groups (COG)-emapper tool. SasA, adaptive-response sensory-kinase; AtoC, signal transduction histidine-protein kinase; S, Rieske iron-sulfur protein; Cyt-*b*, *b*-type cytochrome; MacB and ABC, components of a putative ABC-type transporter complex. **c**, Growth of *S. ovata* in medium amended with yeast extract in the presence of N_2O. Growth of *S. ovata* cultures stalled after one week, presumably owing to substrate depletion. Growth resumed following the addition of H_2, but N_2O reduction was not observed, even after an extended two-month incubation period. Data are mean ± s.d. of triplicate batch incubations. Where error bars are not visible, they are smaller than the symbols or borders. All strain Mol cultures were initiated with $(8.1 \pm 1.7) \times 10^8$ cells.

A putative L-*nos* gene cluster

C-*nosZ* genes occur in conserved gene clusters that encode accessory proteins, which are essential for C-NosZ maturation and function (for example, NosB, NosD, NosF, NosY and NosL)[2,4]. To identify potential *nos* accessory genes related to the single L-*nosZ* on the Sab5 MAG, we performed additional PacBio sequencing, generating a complete genome, including the origin and the terminus of replication. A comparative analysis of 15 *Desulfitobacterium* genomes retrieved from the NCBI database (Supplementary Table 7) revealed that 13 genomes encode clade II C-*nos* clusters with conserved gene content and arrangement (Fig. 3 and Supplementary Note 4), 2 lack C-*nosZ* genes and none carry L-*nosZ*. In anoxic batch reactor incubations, axenic clade II-harbouring *Desulfitobacterium hafniense*, *D. dehalogenans* and *Desulfitobacterium* sp. strain PCE1 reduced N_2O with formate, hydrogen or lactate as electron donor (Supplementary Fig. 5). No *nosB*, *nosD*, *nosF*, *nosY* or *nosL* accessory genes were identified in the genome of *D. nosdiversum* strain Sab5. Instead, the L-*nosZ* gene is surrounded by genes encoding cytochromes and a putative MacB-type transporter complex (a type VII ABC transporter[33]), consisting of a MacB and a putative ABC transport system (Extended Data Fig. 6g,h). The functional roles, if any, of these surrounding genes in N_2O reduction should be the subject of future studies.

NosZ phylogeny and *nos* cluster analysis

L-NosZ and C-NosZ diverge in sequence but have conserved 3D structures. Amino acid identities below 35% and the lack of characteristic accessory proteins highlight the differences between L-NosZ and C-NosZ, and collectively suggest an expansion of the diversity of N_2ORs. Protein sequence-based phylogeny of compiled C-NosZ sequences[34] (n = 173) and L-NosZ sequences (n = 95) confirms that C-NosZ form two independent, monophyletic clades (clade I and clade II), with L-NosZ forming a separate, deep-branching clade with 100% bootstrap support (Extended Data Fig. 7a). Multiple sequence alignments of NosZ reveal a conserved CX_2FCX_3HXEM motif—that is, the Cu_A site, in all L- and C-NosZ (Extended Data Fig. 7b). The characteristic Cu_Z site consisting of seven conserved histidine residues is present in C-NosZ and L-NosZ; however, the motif sequences representing the Cu_Z site differ between C-NosZ and L-NosZ (Extended Data Fig. 7c). Specifically, the DXHH motif that is 100% conserved among C-NosZ is replaced by a GXHH motif in all L-NosZ proteins.

The protein sequence and structure-based phylogenetic analyses support the classification of N_2OR into three distinct groups, clade I and clade II (both C-NosZ type), and clade III, which represents L-NosZ (Fig. 4a). The clade III L-NosZ is phylogenetically more distant from clade I and II C-NosZ than clade I and II C-NosZ are from each other. Clade I and II C-NosZ feature Tat and Sec signal peptides, respectively[2], indicating that they have different mechanisms for translocation into the periplasm. Most L-NosZ sequences (n = 76) possess an N-terminal Sec signal peptide, but others (n = 18, all derived from MAGs) lack a recognizable Sec or Tat signal peptide. The absence of a signal peptide is not unique to L-NosZ, and 24 clade I and 55 clade II C-NosZ proteins encoded on isolate genomes lack signal peptides (Supplementary Table 8). Future studies should explore potential cytoplasmic functions of NosZ proteins that lack a signal peptide.

A BLAST analysis identified five conserved genes neighbouring L-*nosZ* on the *D. nosdiversum* strain Sab5 and the *S. acidovorans* strain Mol genomes. We compared the two putative L-*nosZ* gene clusters to 93 L-*nosZ*-containing contigs in the NCBI non-redundant database and identified 11 putative gene clusters containing L-*nosZ* with at least four shared neighbouring genes (Supplementary Table 9). The majority of L-*nosZ*-carrying contigs (n = 82) shared at least one cytochrome-encoding gene. Genes encoding proteins with predicted regulatory functions, SasA and AtoC, were found only in the genomes of strain Sab5 and strain Mol. None of the 95 contigs containing L-*nosZ*

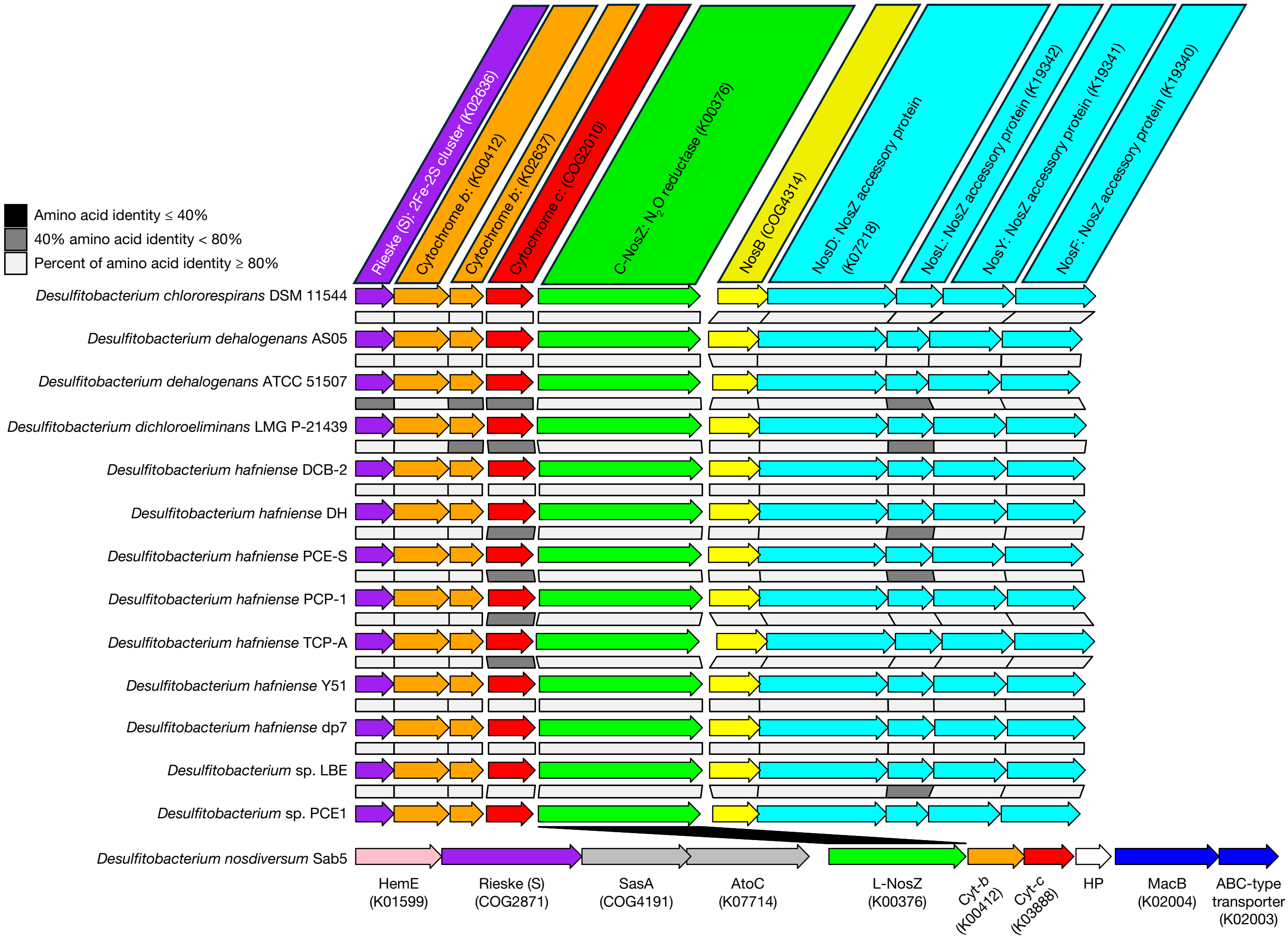

Fig. 3 | The L-*nosZ* locus of *D. nosdiversum* strain Sab5 differs from C-*nos* gene clusters of other *Desulfitobacterium* spp. Clade II C-*nos* gene clusters were identified on 13 out of 15 available *Desulfitobacterium* spp. genomes, and the L-*nosZ* with putative accessory genes was identified on the *D. nosdiversum* strain Sab5 genome. Coloured arrows indicate genes, with the protein designation given on top. Functional annotation was inferred with eggNOG-mapper using the genome mode, with priority given to the KEGG and COG databases. Names of protein-coding genes correspond to: Rieske protein (K02636); Cyt-*b* (K00412 and K02637); Cyt-*c* (COG2010); NosZ (K00376); NosB (hydrophobic protein containing at least four transmembrane-spanning helixes, COG4314); NosD (K07218); NosL (K19342); NosY (K19341); NosF (K19340); MacB (K02004); and ATP-binding cassette transporter (ABC transporter, K02003). Accession numbers in parentheses are KO and COG IDs. The MacB and ABC-type transporter (K02003) may form a 2:2 protein complex based on structural modelling (Extended Data Fig. 6g,h). Shaded boxes separating genes in the different genomes reflect the percentage amino acid identity.

contain any of the characteristic C-*nos* accessory genes (for example, *nosRDFYL* or *nosBDFYL*)[2] (Fig. 4b). Biochemical studies have demonstrated that C-NosZ assembly necessitates accessory proteins (for example, NosDFYL) for copper incorporation[30]. Heterologous expression of C-NosZ in *Escherichia coli* results in a copper-free apo form of the N_2OR that lacks N_2O reduction activity[35]. Most of the 95 genome bins carrying L-*nosZ* represent nearly complete genomes, suggesting that *nos* accessory genes were not missed during assembly. A circular genome and a complete genome are available for *S. acidovorans* strain Mol and *D. nosdiversum* strain Sab5, respectively, and *nos* accessory genes are not present in either case. Therefore, the assembly of active L-NosZ does not depend on accessory proteins encoded on C-*nos* clusters, suggesting alternate mechanisms for copper incorporation and L-NosZ maturation.

L-*nosZ* is underrepresented in genomic databases

To gauge the impact of this previously unknown type of N_2OR on the phylogenomic diversity of N_2O reducers, we built a hidden Markov model (HMM) from 268 N_2OR sequences (Supplementary Fig. 6). These sequences were queried against species-level clustered genome bins (SGBs) compiled in the Genome Taxonomy Database (GTDB) r220[36], a soil genomic catalogue[37] and an ocean genomic catalogue[38]. Analysis using HMMER yielded N_2OR-like sequences, which were filtered by inspecting the Cu_A and Cu_Z motif sequences to establish high confidence in the resulting sequence pool (Supplementary Fig. 6). Subsequent phylogenetic analysis assigned N_2OR sequences to clade I, II and III NosZ groups. Totals of 8,546 (7.5%) GTDB SGBs, 940 (6%) soil SGBs and 378 (4.5%) ocean SGBs possess *nosZ* genes essential for N_2O reduction (Supplementary Tables 10–12). The detection frequencies of *nosZ* in the GTDB SGBs, soil SGBs and ocean SGBs are slightly lower than a previous genome-based prediction, which estimated that around 8% of genomes included in the NCBI database (314 out of 4,135 genomes, accessed November 2012) contain a *nosZ* gene[39].

About 20% of *nosZ*-carrying SGBs, classified mainly as Pseudomonadota (formerly Proteobacteria), possess clade I C-*nosZ*, whereas more than 60% of the SGBs identified contain clade II C-*nosZ*. The SGBs with clade II C-*nosZ* are taxonomically more diverse, spanning

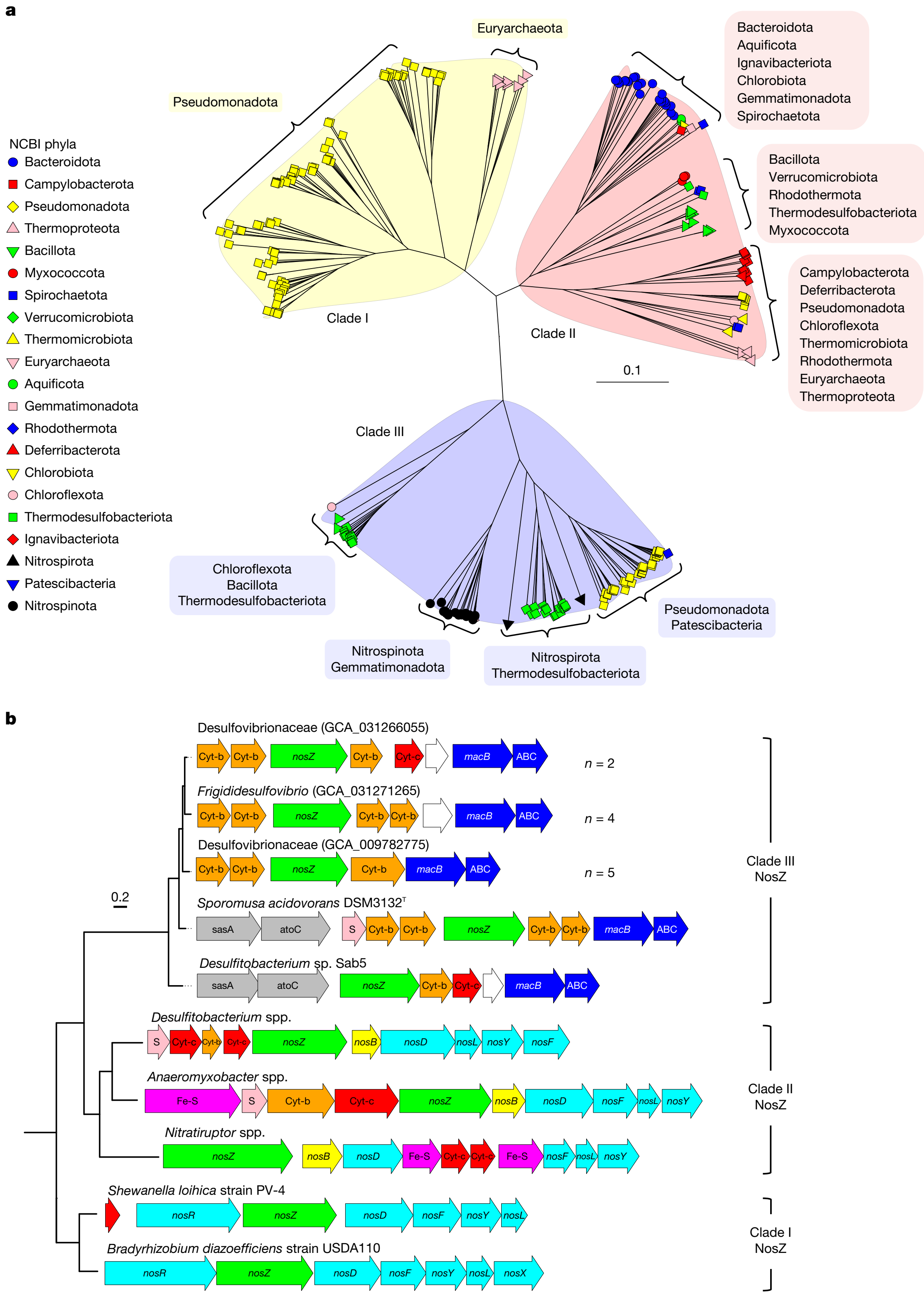

Fig. 4 | Structure- and gene cluster-based analyses separate clade I and clade II C-NosZ from clade III L-NosZ. a, A structure-based phylogenetic tree of 268 AlphaFold 2 models of C-NosZ and L-NosZ was generated using Foldtree and represents the phylogenetic clustering of around 10,000 clade I and II C-NosZ proteins and 101 clade III L-NosZ proteins. NosZ grouping is indicated by yellow (clade I), pink (clade II) and light blue (clade III) shading. The scale bar indicates the mean number of amino acid substitutions per site. Supplementary Fig. 4 shows the homodimeric structures of representative NosZ proteins modelled with AlphaFold 2. **b**, Comparison of representative N_2OR gene clusters from genomes containing C-*nosZ* and L-*nosZ* genes. Several N_2OR gene clusters from genome bins containing L-*nosZ* genes have identical gene arrangements, and three representative clusters are shown. The accessory genes *nosD*, *nosF*, *nosL* and *nosY* are conserved across C-*nos* clusters. The white arrows represent genes of unknown function with no conserved orthologues in known C-*nos* and putative L-*nos* clusters. Genes encoding a MacB and an ABC transporter (forming the putative ABC transport system) are associated exclusively with putative L-*nosZ* (clade III *nosZ*) gene clusters.

55 phyla (Supplementary Figs. 7–9). Genome bins that include L-*nosZ* are found less frequently in SGBs (less than 2%). Because soils are major N_2O sources and sinks, we capitalized on the rapidly expanding soil metagenome datasets for finer-level investigations of the distribution of clade III L-*nosZ* in soil biomes (Extended Data Fig. 8). These efforts reveal that all investigated biomes harbour *nosZ* genes (99 ± 96 counts per Gb), on par with *nrfA* gene abundances (86 ± 69) in the same metagenomic datasets[40]. Phylogenetic placements on the NosZ reference tree showed that metagenomic *nosZ* gene reads grouped at or near the tips, highlighting the robustness of the GraftM approach to classify *nosZ* genes represented by metagenomic short reads (Supplementary Fig. 10). Clade III L-*nosZ* sequences are apparently underrepresented (less than 2% detection frequency) in genomic catalogues, especially those representing soil genomes that remain largely unassembled (and un-binned) in typical sequencing efforts. The investigation of metagenome-derived short-read sequences showed that normalized read counts of clade III L-*nosZ* sequences outnumber clade I C-*nosZ* sequences in 9 out of 12 biomes analysed (Fig. 5), suggesting ubiquitous distribution of L-*nosZ* in soil biomes. The environmental distribution of clade III L-*nosZ* was further resolved at the genome level using stringent metagenome recruitment analysis. The analysis included 179 metagenomes representing a variety of groundwater and marine environments (Supplementary Table 13) and a non-redundant set of 101 genome bins (Supplementary Table 14), each comprising an L-*nosZ*. A subset ($n = 16$) of the 179 metagenomes includes clade III L-NosZ-encoding genome bins, with 0.23% to 1.88% of the metagenome reads recruited to individual reference genomes (Extended Data Fig. 8). The apparent incongruity between clade III L-*nosZ* representation in genome versus metagenome databases indicates inclusion biases[41], highlighting that organisms carrying clade III L-*nosZ* are underrepresented in current genome databases. In support of this incongruity, a recent survey determined that more than 40% of bacterial phylogenetic diversity lacks any genomic representation[42].

Taxa harbouring L-*nosZ*

Using the minimum information about a metagenome-assembled genome (MIMAG) criteria[43], most (87 out of 101) L-*nosZ*-carrying genome bins were identified as medium-to-high quality, 10 were identified as high quality, and 4 were identified as poor quality MAGs. Taxonomic placement of the 101 L-*nosZ*-carrying genome bins against the GTDB database version r220 revealed a broad taxonomic distribution across eight phyla, including Bacillota (3 bins), Chloroflexota (1 bin), Desulfobacterota (29 bins), Nitrospinota (26 bins), Nitrospirota (4 bins), Patescibacteria (1 bin), Pseudomonadota (36 bins) and Spirochaetota (1 bin). The genome bins of Chloroflexota and Spirochaetota are of poor quality and additional sequence data are needed for validation of L-*nosZ* occurrence. Pseudomonadota, Desulfobacterota and Nitrospinota phyla are well represented by L-*nosZ*-carrying genome bins (Supplementary Fig. 11). Taxa with clade I or II C-*nosZ* are common among the Pseudomonadota[2,3], and the observation that many members possess L-*nosZ* suggests an even greater phylogenetic diversity of N_2OR genes in this phylum.

The phylum Desulfobacterota comprises characterized sulfate-reducing[44], fermentative[45] and syntrophic[46] lineages, and our genome-centric investigation found that 1.3% of the available genome bins carry an N_2OR gene (clade I C-*nosZ*, 0.1%; clade II C-*nosZ*, 0.7%; and L-*nosZ*, 0.5%). Recent investigations implicated Desulfobacterota with C-*nosZ* in N cycling in cold marine sediments[47], and the detection of clade III L-*nosZ* expands the N_2OR diversity within this phylum. Cultured members of the class *Nitrospinia* are nitrite-oxidizing representatives of the phylum Nitrospinota[48]. Interestingly, *nxrB* genes encoding nitrite oxidoreductase (which catalyses $NO_2^- \rightarrow NO_3^-$) were not found in the MAGs of uncharacterized Nitrospinota classes[49].

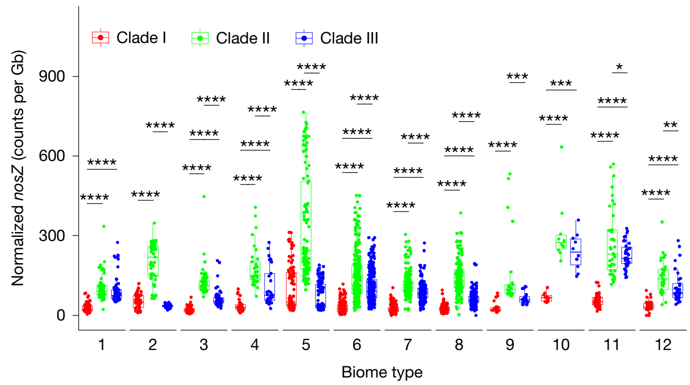

Fig. 5 | Geographic distribution of C-*nosZ* and L-*nosZ* across soil microbiomes. Normalized clade I, clade II and clade III *nosZ* counts (dots) per biome were calculated as the ratio of *nosZ* counts to the total number of bases sequenced ($n = 1{,}053$ metagenomes). Boxes are bounded on the first and third quartiles and horizontal lines represent median values. Whiskers denote 1.5 times the interquartile range. Differences between clade I, clade II and clade III *nosZ* genes were tested using two-sample, two-sided Wilcoxon tests with Benjamini–Hochberg adjustment; $^*P \leq 0.05$, $^{**}P \leq 0.01$, $^{***}P \leq 0.001$, $^{****}P \leq 0.0001$. Biome types are represented by numbers: (1) boreal forest and taiga; (2) croplands; (3) deserts and xeric shrublands; (4) mediterranean forests, woodlands, and scrub; (5) rhizosphere; (6) temperate broadleaf and mixed forests; (7) temperate conifer forests; (8) temperate grasslands savannas and shrublands; (9) tropical and subtropical dry broadleaf forests; (10) tropical and subtropical grasslands, savannas, and shrublands; (11) tundra; and (12) tropical and subtropical moist broadleaf forests.

Instead, members of these uncharacterized classes have the genetic potential to perform nitrate reduction (*napA*) and dissimilatory nitrite reduction to ammonium (*nrfA*). Searching for N_2OR-coding genes in the three genomic catalogues, we found 26 Nitrospinota genome bins, distributed across three uncharacterized Nitrospinota classes (UBA7883, UBA9942 and JACRGQ01), that harbour L-*nosZ*. Previous work showed transcriptional activity of L-*nosZ* of Nitrospinota genome bins GCA_023248495.1 and GCA_023248535.1 in groundwater- and sediment-associated communities at levels resembling those of clade I and II C-*nosZ*[50,51]. The observation of these sequences in metagenomic and/or metatranscriptomic datasets is not unexpected considering the widespread occurrence of the clade III L-NosZ, and our study provides the experimental evidence for the functional role of this protein family. C-*nosZ* sequences were not found in any of the 218 Nitrospinota genome bins investigated, consistent with previous reports[49]. Together, the analysis reveals that most L-*nosZ*-carrying genome bins represent uncharacterized taxa at the class (lowest rank assigned; $n = 48$), order ($n = 7$), family ($n = 9$), genus ($n = 11$) and species ($n = 24$) levels currently assigned non-Latin placeholder names (Supplementary Fig. 11). The distribution of L-*nosZ* across diverse, uncultivated taxa highlights the potential for N_2O reduction among microbial dark matter[41]. Recent findings demonstrate that Nitrospinota representatives dominate nitrite oxidation in oxygen minimum zones (OMZs) in the Eastern Tropical North Pacific[52]. The metagenomic read recruitment analysis did not detect L-*nosZ*-carrying Nitrospinota genome bins in the Eastern Tropical North Pacific OMZ metagenomes ($n = 27$) investigated. Instead, three Pseudomonadota taxa carrying an L-*nosZ* gene were found to be tenfold more abundant than nitrite-oxidizing Nitrospinota populations[52] (Supplementary Fig. 12). Apparently, a previously overlooked N_2O sink also exists in marine OMZs. We further investigated transcription of clade III *nosZ* in the global ocean (Tara Oceans) metatranscriptome dataset[53]. Analysis of the Tara Oceans transcriptomes revealed that a Nitrospinota L-*nosZ* was significantly expressed (1.31 to 17.25 reads per kilobase per million mapped reads (RPKM)) in Indian Ocean OMZs, which was comparable to the expression of clade I and II *nosZ* (1.08 to 32.99 RPKM) in the Arctic and Pacific oceans (Extended Data Fig. 9).

Previous studies have proposed vertical inheritance and horizontal gene transfer between closely related populations (for example, *Desulfitobacterium* and *Bacillus*[54,55]) as drivers of C-*nosZ* evolution[56]. L-*nosZ* occurs in distantly related phyla and the processes that govern L-*nosZ* distribution and evolution are currently unclear. For instance, *S. acidovorans* strain Mol is the only member of the order *Sporomusales* carrying an L-*nosZ* (Supplementary Table 14), suggestive of horizontal acquisition from a distant taxon.

The investigation of the 101 genome bins with L-*nosZ* for potential utilization of alternate electron acceptors reveals 17 with all genes (*narAB*/*narGHIJ*, *nirK*/*nirS*, NOR genes and L-*nosZ*) for complete denitrification ($NO_3^- \rightarrow NO_2^- \rightarrow NO \rightarrow N_2O \rightarrow N_2$) (Extended Data Fig. 10). Thirty-eight percent of genome bins (39 out of 101) with L-*nosZ* harbour genes required for converting NO_2^- to N_2, considerably lower than the percentages of clade I (93%) and clade II (61%) C-*nosZ*-carrying genome bins[4]. A quarter of the genome bins ($n = 25$), including the genomes of *S. acidovorans* strain Mol and *D. nosdiversum* strain Sab5, exclusively harbour L-*nosZ* and no other denitrification pathway genes. Forty-five genome bins harbour genes encoding sulfate adenylyltransferase (*sat*) and/or sulfite reductase (*dsrAB*)[57], suggesting that some L-*nosZ*-carrying organisms can utilize sulfate and/or sulfite as alternate electron acceptor. High (*ccoNOP* and *cydAB*) and/or low (*ctaABCDE*) affinity O_2 reductases[58] are found in 70% (71 out of 101) of L-*nosZ*-carrying genome bins (Extended Data Fig. 10). Genome-based computational model analysis of growth preference[59] reveals that 67.3% (68 out of 101) of the L-*nosZ*-harbouring genome bins represent obligate anaerobes. Both high- and low-affinity oxidases have been found in strict anaerobes[60–62], suggesting roles in O_2 detoxification rather than O_2 respiration. By comparison, only 3.9% (128 out of 3,281) and 16.7% (747 out of 4,476) of clade I and clade II *nosZ*-carrying genome bins from the GTDB database, respectively, represent obligate anaerobes. These findings suggest that L-*nosZ*-carrying bacteria are more likely to inhabit anoxic environments, including OMZs[52,53].

Conclusion and outlook

Current global warming trends are alarming and demand comprehensive understanding of greenhouse gas sources and sinks. Atmospheric N_2O concentrations are increasing, indicating that current N_2O sources outpace N_2O sinks. The predominant N_2O source stems from synthetic N fertilizer applications, and a major goal is to reduce N_2O emissions from agricultural soils and balance N_2O budgets. The major N_2O sink process is microbial reduction to N_2, a process that has been studied in the context of denitrification for more than a century[56]. The more recent discovery that non-denitrifiers commonly harbour C-*nosZ* genes substantially changed our view of N_2O-consuming microbiomes[2,3]. The discovery of L-NosZ, a distinct type of N_2OR, constitutes another significant expansion of the diversity of N_2ORs that catalyse the conversion of the greenhouse gas N_2O to environmentally benign N_2. Taxa carrying L-*nosZ* occur in at least eight phyla and are globally distributed, with the majority representing as yet uncultivated bacteria. Primer sets that have been used to assess the presence, abundance and diversity of C-*nosZ* genes would not amplify L-*nosZ*, highlighting that the N_2O reduction potential in various critical environments warrants re-evaluation. Transcriptome data support that uncultivated Nitrospinota with L-*nosZ* contribute to N_2O consumption in anoxic marine and terrestrial environments. Comprehensive knowledge of N_2ORs will improve the predictive capabilities of Earth system models and strengthen the toolbox for innovative mitigation strategies. Recent work has demonstrated the utility of bioaugmentation for curbing soil N_2O emissions[5,6], and the enlarged diversity of N_2O reducers expands opportunities to potentially re-balance N_2O sources and sinks, thus mitigating further increases of this greenhouse gas in Earth's atmosphere.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-025-09401-4.

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Methods

Soil sampling and microcosms

Tropical soil was collected from the Sabana Tabonuco forest in the Luquillo Experimental Forest (LEF; 18.3° N, 65.80° W) located in the northeastern part of Puerto Rico. Detailed descriptions of the sampling site and soil sampling procedures are available[20,67]. Soil was collected at 5–20 cm depth in August 2018. Replicate microcosms were established with 2 g of homogenized soil in pH 4.5 medium amened with 5 mM lactate, and were maintained under low and high-N_2O feeding regimens, as described[20]. Physicochemical properties and baseline metagenome data of the Sabana soil have been reported[20,67].

Sab enrichment cultures

Basal salt medium used for transfer cultures was prepared using the Hungate technique as described[68]. The basal salt medium contained (in g l^{-1}): NaCl 1.0, $MgCl_2{\cdot}6H_2O$ 0.5, KH_2PO_4 7.0, NH_4Cl 0.3, KCl 0.3, $CaCl_2{\cdot}2H_2O$ 0.015, L-cysteine 0.031, TSB 0.02, 1 ml trace element solution, 1 ml Se/Wo solution, and 0.25 ml resazurin solution (0.1% w/w). The trace element solution contains (per litre): $FeCl_2{\cdot}4H_2O$ 1.5 g, $CoCl_2{\cdot}6H_2O$ 0.19 g, $MnCl_2{\cdot}4H_2O$ 0.1 g, $ZnCl_2$ 70 mg, H_3BO_3 6 mg, $Na_2MoO_4{\cdot}2H_2O$ 36 mg, $CuCl_2{\cdot}2H_2O$ 2 mg, and 10 ml HCl (25% solution, w/w). The Se/Wo solution consists of (mg l^{-1}): $Na_2SeO_3{\cdot}5H_2O$ 6, $NaWO_4{\cdot}2H_2O$ 8, and NaOH 500. The final medium pH following autoclaving ranged between 4.2 to 4.4 and no pH adjustment was conducted. A volume of 0.1 ml filter-sterilized Wolin vitamin solution was added to 100 ml medium with 1 ml syringes prior to inoculation. The Wolin vitamin stock solution (1,000-fold concentrated) comprises the following (all as mg l^{-1}): biotin 20, folic acid 20, pyridoxine hydrochloride 100, riboflavin 50, thiamine 50, nicotinic acid 50, pantothenic acid 50, vitamin B_{12} 1, p-aminobenzoic acid 50, thioctic acid 50. Volumes of 10 ml (4.16 mM, nominal) H_2 and N_2O were injected into vessels and equilibrated overnight prior to inoculation from a Sabana soil microcosm with N_2O reduction activity. Fumarate (1 M autoclaved stock solution), rather than lactate, was added to a final concentration of 5 mM. Initial attempts to obtain solid-free enrichment cultures from a Sabana soil microcosm with 5 mM lactate as carbon source and electron donor were unsuccessful (that is, no N_2O reduction activity was observed). Subsequent efforts tested combinations of different carbon substrates, H_2 and TSB, and a combination of 5 mM fumarate, 4.16 mM (nominal) H_2, and 0.02 g l^{-1} TSB supported N_2O reduction in transfer cultures. Cell morphologies in transfer cultures were regularly examined using light microscopy (Zeiss Axioscope with AxioVision software, Carl Zeiss Microscopy).

Cultivation of clade III L-*nosZ*-carrying *S. acidovorans* strain Mol

An axenic culture of *S. acidovorans* strain Mol, characterized as a strict anaerobe[28], was obtained from the German Collection of Microorganisms and Cell Cultures (DSMZ) (https://www.dsmz.de/collection/catalogue/details/culture/DSM-3132). *S. ovata* DSM2662^T was kindly provided by C. Liu. The *Sporomusa* spp. cultures were cultivated in 160 ml glass serum bottles containing 100 ml 30 mM bicarbonate-buffered basal salt medium. The basal salt medium is identical to that described for cultivating the Sab enrichment culture except that the amount of KH_2PO_4 was 0.2 g l^{-1} and TSB was omitted. *S. acidovorans* strain Mol requires yeast extract for growth and both *Sporomusa* spp. cultures received 1 g l^{-1} yeast extract[28]. The pH of medium was adjusted to ~7 with CO_2. A volume of 1 ml N_2O was added to the cultivation vessels followed by an overnight equilibration prior to inoculation. To test if H_2 serves as an electron donor for N_2O reduction, *S. acidovorans* strain Mol cultures received 5 ml of H_2 after N_2O reduction had stalled, presumably due to the depletion of yeast extract-derived reducing equivalents. Additional *S. acidovorans* strain Mol cultures received varying volumes (0–10 ml) of N_2O to investigate if N_2O reduction contributes to growth. Negative control incubations included *S. acidovorans* strain Mol cultures with 2 ml of N_2O and 6 ml of acetylene, an inhibitor of N_2OR activity, and cultures that received heat-killed (autoclaved) inoculum.

Cultivation of clade II C-*nosZ*-carrying *Desulfitobacterium* species

Axenic *Desulfitobacterium* cultures (*D. dehalogenans*, *Desulfitobacterium* sp. strain PCE1 and *D. hafniense*) were grown in 160 ml glass serum bottles containing 100 ml of anoxic, reduced, bicarbonate-buffered basal salt medium[67,69]. A volume of 1 ml N_2O was added to the serum bottles followed by an overnight equilibration period prior to inoculation. Formate (30 mM) was added as electron donor from an anoxic, autoclaved stock solution.

Metagenomic workflow and phylogenomic analysis

To identify the population(s) responsible for N_2O consumption in the Sab enrichment culture, genomic DNA was isolated from cultures grown with fumarate (5 mM), TSB (0.02 g l^{-1}), and H_2 (4.16 mM nominal) in the presence (10 ml, 4.16 mM nominal) and absence of N_2O. Cell suspension samples were harvested following a four-day incubation period, after which fumarate and N_2O had been completely consumed. Sequencing libraries were prepared using the Illumina DNA Prep kit (formerly Nextera DNA Flex Library Prep). Metagenome sequencing used the Illumina NovaSeq 6000 platform available in the University of Tennessee, Knoxville (UT) Genomics Core. Metagenomic short reads were processed using the nf-core/mag pipeline[70] on the UT High Performance and Scientific Computing cluster. Individual steps of the nf-core/mag pipeline include evaluation of short read quality with FastQC v.0.11.9 (ref. 71), followed by quality filtering and Illumina adapter removal using fastp v.0.20.1 (ref. 72). Short reads mapped to the PhiX genome (GCA_002596845.1, ASM259684v1) were removed with Bowtie2 v.2.4.2 (ref. 73). The clean short reads of the two metagenomes (that is, from Sab enrichment cultures grown with and without N_2O) were pooled and assembled with Megahit2 v.1.2.9 (ref. 74). Genome binning of assembled contigs used MetaBAT2 v.2.15 (ref. 75) and MaxBin2 v.2.2.7 (ref. 76), considering read coverage, coverage variance, and tetranucleotide frequencies. The output MAGs were refined and dereplicated using DASTool v.1.1.6 through a single-copy gene (SCG) scoring strategy coupled to an iterative bin dereplication procedure that produces the highest-scoring set of non-redundant bins (in terms of SCG completeness/contamination) from input bins generated with different binning approaches[77]. Completeness, size, and contamination levels of the final MAGs were estimated with CheckM2 v.1.0.2 (ref. 78). MAGs that met or exceeded medium quality (completeness >50% and contamination <10%) on the basis of the MIMAG standard[43] were included in further analyses. The taxonomic placements of the MAGs with medium quality or higher were inferred on the basis of GTDB-TK v.2.3.2 (ref. 79).

Phylogenetic reconstruction of *Desulfitobacteriaceae* species included 30 genomes obtained from the NCBI database (accessed May 2024), and the *D. nosdiversum* strain Sab5 genome constructed from metagenome datasets of Sab enrichment cultures. The GTDB-TK v.2.3.2 toolkit[79] was used to identify, align, and concatenate 120 conserved bacterial marker genes in the 31 *Desulfitobacteriaceae* genomes. A maximum likelihood tree topology was computed with RaxML-NG v.1.2.2 (ref. 80) using the best evolutionary models (LG+I+G4+F) inferred by ModelTest-NG v.0.1.7 (ref. 81). The pairwise average amino acid identity (AAI) of single-copy proteins between *D. nosdiversum* strain Sab5 and related *Desulfitobacteriaceae* genomes (n = 30) was computed using fastAAI[82]. Visualization of the genome maximum likelihood tree topology and associated AAI data were conducted using the ggtree package v.3.15.0 (ref. 83) implemented in the R platform.

Circular genome binning and analysis

A draft genome of *S. acidovorans* strain Mol comprising 126 contigs has been reported[84]. To reduce fragmentation, available raw sequence reads[84] were assembled with Flye v.2.9.5 (ref. 85) to generate a single circular contig of 5.95 Mb. Attempts to close this single circular contig with Circlator v.1.5.5 (ref. 86) were unsuccessful. Polypolish v.0.6.0 (ref. 87) was used to correct the single contig with Illumina reads. This Polypolish step was iterated 10 times with a custom script (Polypolish_iteration.py; https://github.com/utguang/Clade3-NosZ), producing a final assembly of 5.96 Mb (deposited in figshare; https://doi.org/10.6084/m9.figshare.29260736). FastANI v.1.34 (ref. 88) was used to compute an average nucleotide identity (ANI) of 99.92%.

Assembly and binning of metagenomic sequences generated from the Sab enrichment culture grown with N_2O yielded a draft genome of *D. nosdiversum* strain Sab5 comprising 105 contigs. A single-molecular real-time (SMRT) bell library was prepared and sequenced following the PacBio manufacturer's protocol (NovoGene). PacBio reads were corrected using Canu (v.2.2)[89] and assembled with metaFlye[90] to generate a single circular contig of 4.26 Mb, which was closed with Circlator v.1.5.5 (ref. 86). The closed circular genome was refined using the raw Illumina reads in ten iterations with Polypolish v.0.6.0 (ref. 87). The ANI of the closed and the fragmented genomes of *D. nosdiversum* strain Sab5 exceeded 98.99%.

Functional annotation and sequence analysis

Functional annotations of the Sab enrichment culture-derived co-assembly and the MAGs were performed using eggNOG-mapper v.2.1.6 (ref. 91) with flags -m diamond --itype genome --decorate_gff yes --genepred prodigal --pident 40. Initial functional annotation of the *D. nosdiversum* strain Sab5 genome and the co-assembly did not identify putative genes encoding N_2OR. Thus, the flag --pident was relaxed to 35 and 30 in further functional annotations, and a putative gene (L-*nosZ*) encoding N_2OR was identified in the co-assembly and the strain Sab5 genome when --pident was set to 30. The L-*nosZ* gene found on the strain Sab5 genome was translated into protein sequence using Prodigal v.2.6.3 (ref. 92) and queried against the NCBI non-redundant database (May 2024). Sequence hits with E-values $<10^{-5}$ were inspected for motif sequences (CXXFCXXXHXEM, DXHH, PHG, GPLH and EPH) characteristic for clade I and clade II C-NosZ. The motif inspection criterion was relaxed to four motif sequences (DXHH was excluded). A total of 94 protein sequence were considered putative L-NosZ following motif inspection.

Comparative analysis of *Desulftobacterium* genomes

The genomic coordinate information of *nos* gene clusters of characterized *Desulfitobacterium* genomes was extracted using an in-house script (gff_extract.py; https://github.com/utguang/Clade3-NosZ). The genes neighbouring *nosZ* on the *D. nosdiversum* strain Sab5 genome were extracted using the gff_extract.py script. The gff files resulting from gff_extract.py contained *nos* cluster coordinate information, which was visualized using the gggenome package[93] implemented in R.

Metagenome recruitment analysis

Recruitment analysis of metagenomic sequence reads of the Sabana soil microcosms and Sab enrichment cultures was performed with CoverM v.0.7.0 (ref. 94), using Sab enrichment culture-derived MAGs as reference genomes, with the flags minimap2-sr –min-read-aligned-percent 50 –min-read-percent-identity 0.95 --min-covered-fraction 0.99. The flag --min-covered-fraction 0.99 secured stringent coverage computation. Relative abundances of Sab enrichment culture-derived MAGs were visualized using the ggplot2 package[95] in R. Coverage evenness of MAGs in metagenomes was evaluated via blasting short reads against genomes with Bowtie2, and the resulting sorted alignment file was visualized using RecruitPlotEasy[96] embedded in a custom script (recruitplot.py; https://github.com/utguang/Clade3-NosZ). A depth-versus-position graph summarizing the *D. nosdiversum* strain Sab5 MAG coverage in Sab enrichment culture metagenomes was created with the ggplot2 package.

Structural modelling analysis

Amino acids corresponding to signal peptides, predicted with SignalP6.0 (ref. 97), were trimmed from 173 C-NosZ and 95 L-NosZ sequences. Functional C-NosZ are homodimers; thus, the C-NosZ and L-NosZ sequences were dimerized with a custom Python script (Monomer2Dimer.py; https://github.com/utguang/Clade3-NosZ). AlphaFold 2 models of homodimeric C-NosZ and L-NosZ were generated using LocalColabFold[98] with default settings. Crystal structures of clade I C-NosZ (PDB entries: 1FWX, 1QNI, 3SBQ, 3SBR and 9F8X) were used as templates. Models were ranked according to an established ranking approach[99] (model confidence = 0.2 × pTM + 0.8 × *ipTM) and all model confidence values exceeded 0.85. Representative models were visualized with ChimeraX v.1.9 (ref. 100).

An AlphaFold 2 model of the putative MacB-type transporter, encoded on genes neighbouring L-*nosZ* in the *D. nosdiversum* strain Sab5 genome, was generated using AlphaFold 2 multimer v.3 (ref. 99) and implemented in a local ColabFold installation. A crystal structure of the MacB-type transporter from *Acinetobacter baumannii* (PDB entry: 5WS4) was used as a template.

L-NosZ was further modelled with AlphaFold 3 (https://alphafoldserver.com), which enables various ligands and ions to be included in the models[32]. Signal peptides predicted with SignalP6.0 (ref. 97) were removed prior to modelling. Cu ions (n = 12–20) were included to predict their coordination with histidine residues in the Cu_Z site. The inclusion of 12 Cu ions resulted in models with 3 of the expected 4 Cu ions present in the 2 Cu_Z sites. Models were then generated by 'saturating' with additional Cu ions. All 4 Cu sites (2 Cu_A and 2 Cu_Z sites) were occupied when modelling included 20 Cu ions. The random seed was set to '1234567' aimed at securing modelling reproducibility. As observed for the AlphaFold 2 models of L-NosZ, the AlphaFold 3 models demonstrated that one of the seven Cu-coordinating histidine residues originates from a β-strand in L-NosZ, while the superposed histidine residue in C-NosZ originates from a different β-strand. Although the position and relative orientation of this histidine residue differ, the models predict that coordination of copper occurs in a similar manner. Specifically, H129 and H130 in the PsNosZ structure originate from the same β-strand but H178 is on a different strand. In comparison, H116 and H117 in the *D. nosdiversum* strain Sab5 NosZ model are from the same β-strand but H62 (corresponding to H129 in PsNosZ) is from a different β-strand. Similarly, H125 and H126 in the *S. acidovorans* strain Mol NosZ model originate from the same β-strand but H71 (corresponding to H129 in PsNosZ) is from a different β-strand.

Quantitative PCR

To determine the growth yield of L-*nosZ*-carrying *D. nosdiversum* strain Sab5 with N_2O as electron acceptor in the Sab enrichment culture, cell suspension samples were collected following the consumption of 0, 41.6, 83.2, 208, and 416 µmol of N_2O. SYBR Green qPCR assays targeting the L-*nosZ* genes of *D. nosdiversum* strain Sab5 and *S. acidovorans* strain Mol were designed using Geneious Prime (Supplementary Table 15). The primers' specificity was examined by in silico analysis using the Primer-BLAST tool, and experimentally confirmed using synthesized linear DNA fragments (IDT) of the L-*nosZ* genes of strain Sab5 and strain Mol. For enumeration of L-*nosZ* genes, 25-µl qPCR tubes received 10 µl 1× Power SYBR Green, 9.88 µl UltraPure nuclease-free water (Invitrogen), 300 nM of each primer, and 2 µl template DNA. All qPCR assays were performed using an Applied Biosystems ViiA 7 system (Applied Biosystems) with the following amplification conditions: 2 min at 50°C and 10 min at 95 °C, followed by 40 cycles of 15 s at 95 °C, and 1 min at 60 °C. The standard curves generated with 10-fold serial dilutions of the linear DNA fragments carrying a complete sequence of the L-*nosZ* gene of strain Sab5 (1,599 bp) or of strain Mol (1,635 bp) had slopes of −3.471 and −3.414, y intercepts of 36.431 and 34.076, R^2 values of 0.998 and 0.998, and qPCR amplification efficiencies of 94.1% and 96.3%, respectively.

The linear range spanned 8.54 to 8.54×10^7 and 7 to 7×10^7 gene copies per assay tube. The computational analysis indicated that both strain Sab5 and strain Mol carry a single L-*nosZ* gene, indicating that the enumeration of L-*nosZ* genes directly estimates cell abundances.

(Meta)proteomics analysis

(Meta)proteomics analysis was performed with cells collected from the Sab enrichment culture grown with N_2O. Comparative proteomics analyses were performed with axenic *S. acidovorans* strain Mol cells grown with and without N_2O. Biomass was harvested when at least two-thirds of the initial N_2O was consumed in Sab enrichment cultures and in axenic *S. acidovorans* strain Mol cultures. Suspension samples (50 ml) were withdrawn, the biomass collected by centrifugation, and the pellets immediately frozen at −80 °C. Protein extraction was conducted using a protein aggregation capture method[101], followed by digestion with a protein to trypsin ratio of 1:75 (w/w). The digested samples were analysed using a Vanquish UHPLC system connected to an Orbitrap Q-Exactive Plus mass spectrometer (Thermo Fisher Scientific) utilizing a 1D liquid chromatography–tandem mass spectrometry approach. Two micrograms of each sample were processed, with peptides trapped on a 5 µm Kinetex Phenomenex trapping column before separation on a 1.7 µm Kinetex Phenomenex analytical column. Desalting was carried out for 30 min at 2 µl min^{-1} using solvent A (0.1% formic acid in 5% acetonitrile). Analytical separation and elution were achieved through an increasing gradient ranging from 0–30% solvent B (0.1% formic acid in 70% acetonitrile) over 185 min, followed by a column wash. Peptides were analysed in data-dependent mode and sequenced throughout the first 240 min of the total 275 min analysis time. Full-scan mass spectra were collected in the m/z 300–1,500 range at a resolution of 70,000 (full-width at half maximum). The twenty most intense precursor ions were selected for MS/MS with an isolation window set to 1.8 m/z and a dynamic exclusion window of 30 s. Peptide fragmentation analysis was conducted using Proteome Discoverer (v.2.3.0.523, Thermo Scientific) employing the MS Amanda algorithm (v.2.0) and a target-decoy approach against the corresponding (meta) genome and a common protein contaminants database[102]. Peptide filters included fully tryptic peptides, up to two missed cleavages, and a minimum peptide length of five amino acids. A match between runs was performed on triplicate sets, with a peptide false discovery rate threshold of 1%. Extracted ion chromatograms of peptides were used for abundance quantification, with protein quantification calculated by summing unique peptide abundances.

Protein sequence- and structure-based phylogeny reconstruction of C-NosZ and L-NosZ

Protein sequence-based phylogeny reconstruction of C-NosZ and L-NosZ included clade I (n = 105) and clade II (n = 68) C-NosZ precompiled in a ROCker model (http://enve-omics.ce.gatech.edu/rocker/rocker/models/NosZ/NosZ.ref.fasta), and 95 clade III L-NosZ identified in this study. The 173 C-NosZ and 95 L-NosZ sequences were collectively aligned with MAFFT using the --auto parameter. The NosZ alignments were trimmed with trimAl v.1.4.1 with the -automated1 flag[103]. The trimmed alignments were used to create a maximum likelihood protein tree using RaxML-NG v.1.2.2 with flags --all --tree 10 --bs-tree 1000. In addition, multiple sequence alignments of 105 clade I C-NosZ, 68 clade II C-NosZ and 95 L-NosZ were performed independently aimed at identifying discriminating features of C- and L-NosZ. Multiple sequence alignments were visualized in Geneious and inspected for the conserved NosZ motifs (Cu_A and Cu_Z sites). Sequence logos representing the conserved motifs of NosZ were created using the ggmsa package[104] in R. Structure-based phylogeny reconstruction consisted of 268 structure models of C- and L-NosZ and was performed using Foldtree[105] containerized in Snakemake[106]. Protein sequence- and structure-based tree topologies and associated data were visualized using the ggtree package in R.

Searching for conserved genes neighbouring L-*nosZ*

A de novo search of putative accessory genes neighbouring L-*nosZ* was performed using cblaster v 1.3.20 (ref. 107) with flags -g 1000 -mh 5. The L-nosZ-containing contig used to assemble the *D. nosdiversum* strain Sab5 genome was iteratively blasted against the L-*nosZ*-containing contigs of MAGs (n = 94). This iterative cblaster search yielded seven genes neighbouring L-*nosZ* on the strain Sab5 MAG, which were considered accessory genes putatively involved in L-NosZ function. These putative accessory genes were blasted against the 94 L-*nosZ* containing contigs using the same cblaster flags. The gbk files resulting from cblaster v.1.3.20 were visualized with clinker v.0.0.31 (ref. 108) and gggenomes[93] in R.

Searching for C-*nosZ* and L-*nosZ* in genomic catalogues

An HMM built from the 268 C-NosZ and L-NosZ sequences was used to search public genomic catalogues (GTDB[36], soil[37] and ocean[38] genomic catalogues). SGBs of the GTDB (n = 113,104), soil (n = 16,530) and ocean (n = 8,466) genomic catalogues were obtained from the respective sources. Open reading frames of SGBs were predicted using Prodigal v.2.6.3 and translated into protein sequences. Protein sequence hits with E-values <10^{-5} were filtered by length and motif sequences to secure high confidence of the ultimate NosZ sequences. Multiple sequence alignments of NosZ sequences derived from the GTDB (n = 8,802), soil (n = 1,005) and ocean (n = 382) genomic catalogues were conducted with MAFFT[109] with --auto flag. Fasttree v.2.1.10 (ref. 110) was used to build approximate maximum likelihood trees using default flags. Taxonomic information of SGBs were retrieved from the respective sources of the genomic catalogues. Phylogenetic trees and taxonomic affiliations were compiled using itol.toolkit package v.1.1.9 (ref. 111) in R and visualized using the online tool iTOL v.6 (ref. 112).

Distribution of clade III L-NosZ-encoding genome bins

A total of 102 genome bins carrying L-*nosZ* were compiled and used as reference genomes for environmental metagenome recruitment analysis. A total of 179 metagenomes were obtained with the nf-core/fetchng pipeline[113] by tracking the data accession numbers (Supplementary Table 14). Relative abundances of the MAGs and genomes in each metagenome were computed using CoverM with flags --min-read-aligned-percent 50 --min-read-percent-identity 0.95 --min-covered-fraction 0.99.

nosZ transcripts in the Tara Oceans metatranscriptomic dataset

To investigate C- and L-*nosZ* expression in the oceans, assembled transcripts and relevant expression profiles were obtained from the Tara Oceans dataset (http://www.ocean-microbiome.org)[53]. An HMM containing 268 C- and L-NosZ sequences was queried against the assembled transcripts. Transcripts with E-values <10^{-5} were filtered by length and motif sequences to secure high confidence in the analysis. This workflow resulted in a total of 400 non-identical *nosZ* transcripts, of which eight exceeded the expression level threshold of 1 RPKM.

Construction of a GraftM package targeting *nosZ* gene fragments in metagenomes

The presence of *nosZ* fragments in metagenomes was assessed with GraftM v.0.15.0 (ref. 114), a gene-centric and phylogeny-informed classifying tool. GraftM classifies metagenomic gene fragments in a two-step approach; first, the tool identifies the gene fragments using HMMER v.3.4 (ref. 115) with a reference, and then places them into a pre-constructed phylogenetic tree using pplacer v.1.1 (ref. 116). The phylogenetic placement validates sequence differences and infers clade affiliations (clades I, II and III) of metagenomic gene fragments. A reference package for GraftM was built using the trimmed alignment and associated phylogeny, comprising precompiled clade I NosZ (n = 105), clade II NosZ (n = 68), and clade III NosZ (n = 95) sequences.

Tree statistics required for metagenomic gene fragment assignments were generated using RaxML-NG v.1.2.2.

A collection of 1,035 globally distributed rhizosphere and soil metagenomes[117], and 18 metagenome datasets generated from agricultural soil experiencing regenerative farming practices (Caney Fork Farms) were used to survey the abundance and diversity of *nosZ* communities across microbiomes (Supplementary Table 16). The 1,035 soil metagenomes were compiled in a Zenodo database[40]. The associated metadata of the metagenomes, including accession numbers, were downloaded from the Zenodo database (https://zenodo.org/records/8026657). Metagenomes available in the NCBI and EMBI databases were downloaded using the nf-core/fetchng pipeline. Soil metagenomes obtained by the National Ecological Observational Network (NEON) were downloaded via directing the url link with the wget command (Supplementary Table 17). The abundance and diversity of *nosZ* gene fragments in metagenomes were assessed with GraftM; the metagenome reads corresponding to *nosZ* gene fragments were identified by mining each metagenome with the HMM generated from the NosZ sequence alignment, and candidate metagenome reads were then mapped to the branches of the tree by phylogenetic placement. GraftM was run on the forward reads of each metagenome with default parameters. The resulting jplace files were used to assign *nosZ* gene fragments to clades I, II and III with R packages ggtree and treeio; *nosZ* gene fragments placed under nodes of 249, 428 and 338 are classified as clades I, II and III *nosZ*, respectively. For each metagenome, the *nosZ* gene counts were normalized by the number of bases sequenced to account for differences in sequencing depth (normalized counts). The ratios of clades I, II and III *nosZ* genes were computed with the R package ggscatterpie.

Analytical procedures

N_2O, CO_2, and H_2 were analysed by manually injecting 100 µl headspace samples equilibrated to atmospheric pressure into an Agilent 3000 A Micro-Gas Chromatograph equipped with Plot Q and molecular sieve columns coupled with a thermal conductivity detector[118]. Gas concentrations were calculated by comparing the signals to external calibration curves using in the EZ IQ software (v.3.3.2 SP2, INFICON) considering pressure changes during the incubation. Aqueous concentrations (µM) were calculated from the headspace partial pressures based on reported Henry's law constants[119] for N_2O (2.4×10^{-4}), H_2 (7.8×10^{-6}) and CO_2 (3.3×10^{-4}) mol (m^3 Pa)$^{-1}$ according to

$$H^{\mathrm{cp}}RT = \frac{C_a}{C_g} \quad (1)$$

where H^{cp} is the Henry's law constant[119], R is the universal gas constant, T is the temperature, C_g is the headspace gas concentration (µM) and C_a is the liquid phase (dissolved) concentration. Five-point standard curves for N_2O, CO_2 and H_2 spanned concentration ranges from 0 to 125,000 parts per million by volume. Fumarate, acetate, propionate, succinate and formate were measured with an Agilent 1200 Series high-performance liquid chromatography (HPLC) system[118], and the data processed with the Open Lab ChemStation software. pH was measured in 0.4 ml samples of culture supernatant following removal of cells by centrifugation with a calibrated pH electrode.

Description of *Desulfitobacterium nosdiversum* sp. nov

Desulfitobacterium nosdiversum sp. nov

Etymology. *nosdiversum* (nos.di.ver'sum. N.L. pref. nos- arbitrary prefix referring to *nosZ* genes; L. neut. part. adj. diversum diverse or distinct; N.L. neut. part. adj. nosdiversum referring to distinct *nosZ* genes).
Holotype. The genome has been deposited in the SeqCode registry (https://seqco.de/r:eml1oxac).
Locality. Sabana soil collected in the Luquillo Experimental Forest, Puerto Rico, USA[20,67].
Diagnosis. *nosdiversum* strain Sab5 was enriched from Sabana soil on the basis of its ability to grow with N_2O as electron acceptor at pH 4.5.

Statistics and reproducibility

All experiments were repeated independently at least three times with similar results. Quantitative assays (such as metabolite measurements and gene abundances) were conducted with biological replicates ($n = 3$ per experimental condition). For each biological sample, three technical replicate qPCR assays were performed. Microscopic observations were reproduced across three independent sample preparations. Additional information about replicates is provided in the figure legends or the methods subsections.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The Sab enrichment culture metagenomic datasets (both Illumina and PacBio data) and the resulting closed genome of *D. nosdiversum* strain Sab5 are deposited under NCBI BioProject ID PRJNA1176277. The Sabana microcosm metagenomes are publicly available in NCBI Bioproject ID PRJNA901179. Illumina and PacBio datasets used for construction of a circular genome of *S. acidovorans* strain Mol are publicly available under NCBI BioProject ID PRJNA310710. Genome bins analysed in this study are from the Genome Taxonomy Database (https://gtdb.ecogenomic.org), a global soil genome bins database (https://microbma.github.io/project/SMAG.html) and an ocean genome bins database (https://doi.org/10.6084/m9.figshare.c.5564844.v1). Other genomes and raw metagenomic sequence reads analysed in this study are publicly available in the NCBI database. Lists of the datasets are summarized in Supplementary Tables 7, 8, 10–14, 16 and 17. The mass spectrometry proteomics raw data and sequence files are available at the ProteomeXchange Consortium via the MassIVE database with the unique dataset identifier MSV000096274. The metaproteomic search results are publicly available through the MassIVE database (ftp://massive-ftp.ucsd.edu/v07/MSV000096274/).

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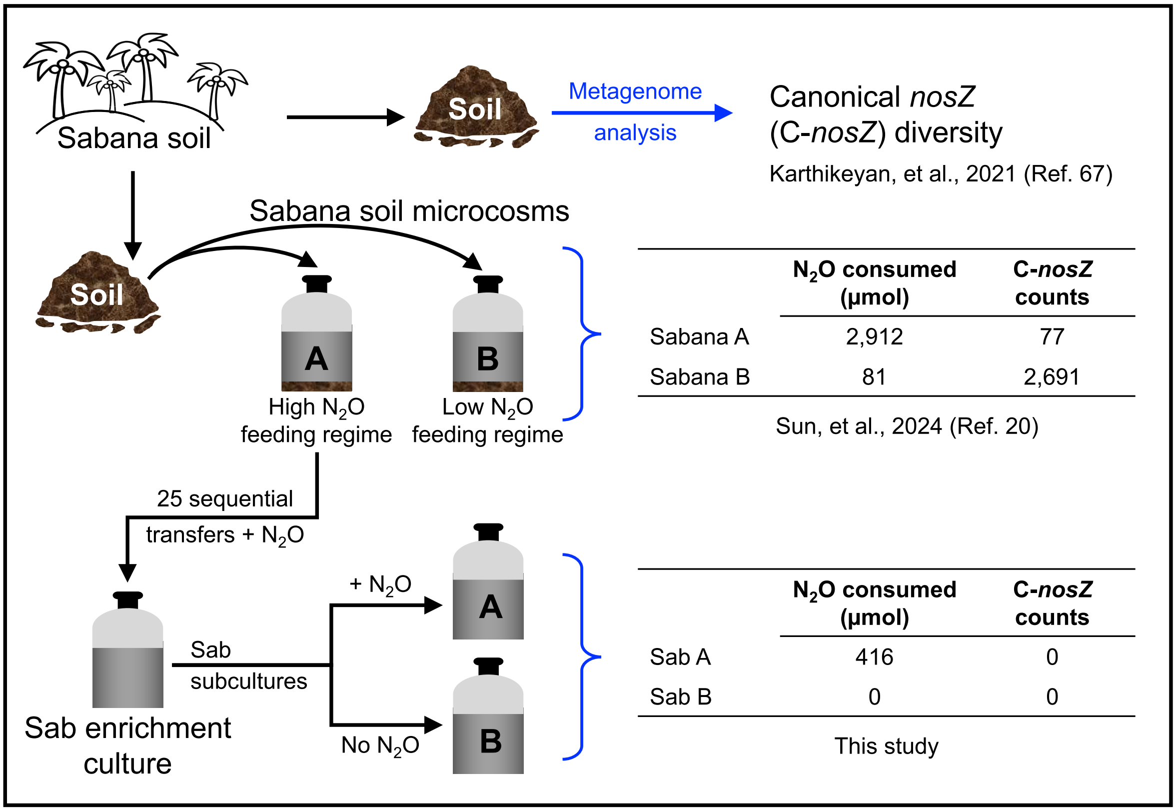
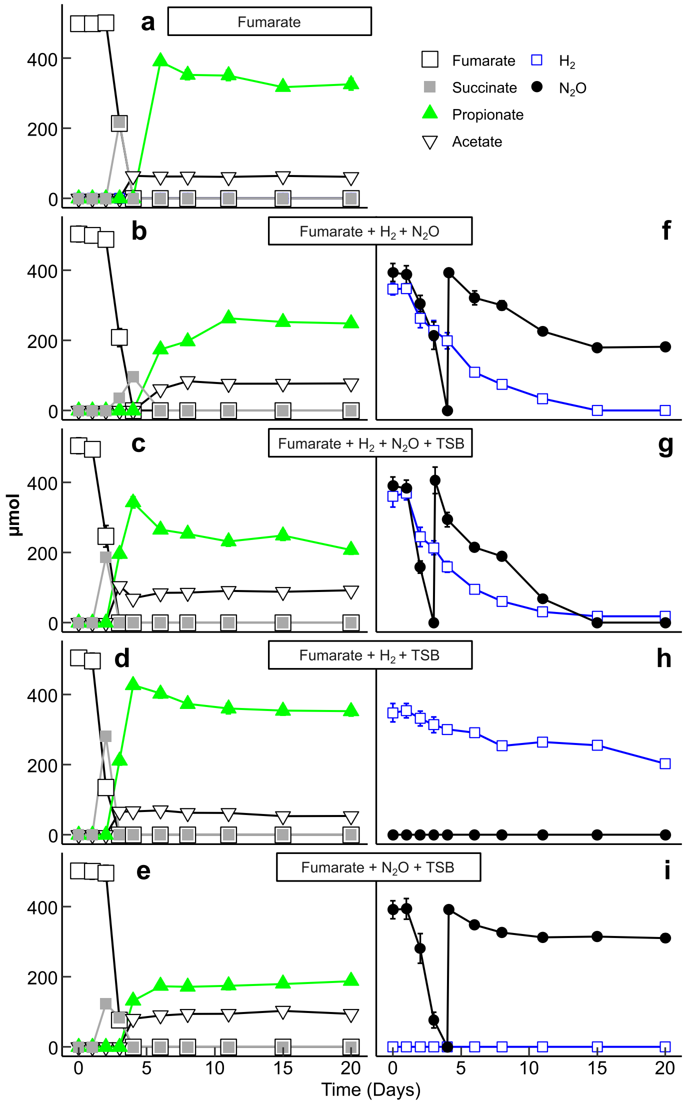
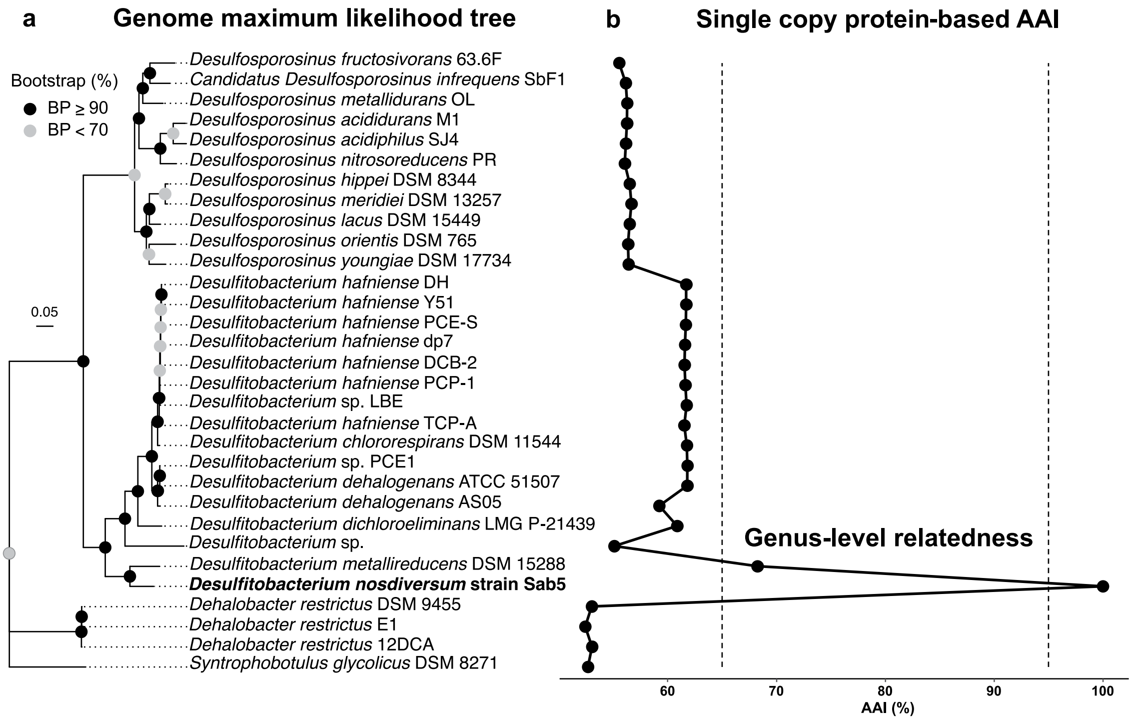


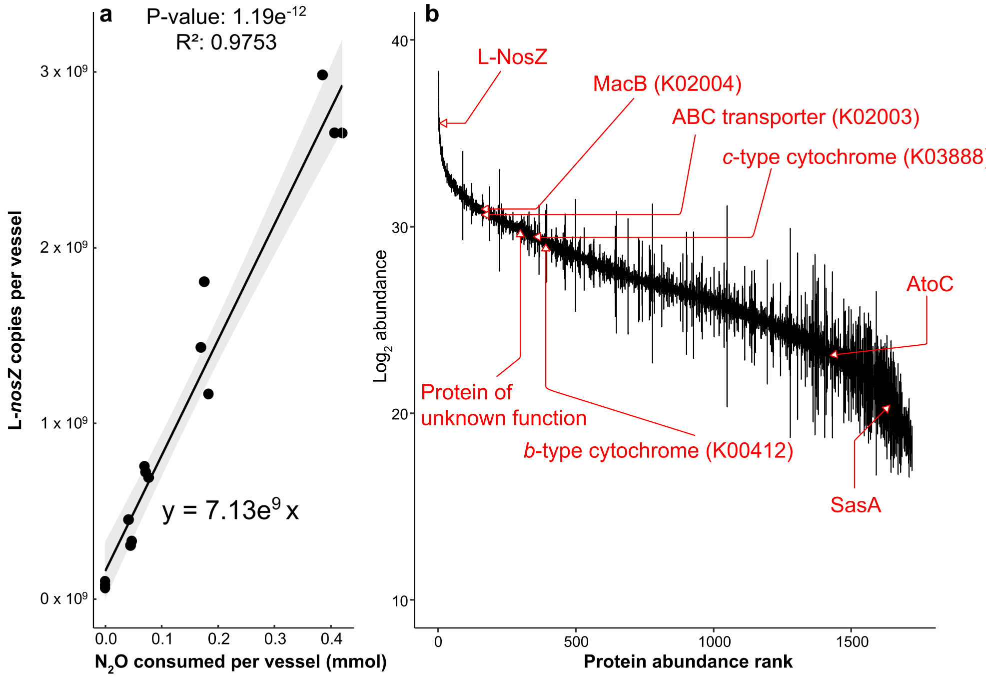
Extended Data Fig. 1 | Schematic of the workflow leading from Sabana forest soil to soil microcosms to solid-free Sab enrichment cultures. Metagenome analysis of the original soil suggested tropical soils harbor diverse C-*nosZ* genes[67]. Efforts aimed at unraveling the microbiology of N_2O consumption in Sabana soil microcosms experiencing different N_2O feeding regimes revealed an incongruity between N_2O consumption and *nosZ* abundance[20]. Twenty-five consecutive transfers starting with the Sabana soil microcosm experiencing a high N_2O feeding regime yielded a Sab enrichment culture with robust N_2O consumption activity. The Sab enrichment culture was incubated in 160 mL glass serum bottles containing 100 mL of basal salt medium amended with 0.02 g L^{-1} TSB, 5 mM fumarate, 10 mL (4.16 mM nominal; 416 µmol) H_2, and 10 mL (4.16 mM nominal; 416 µmol) N_2O. While the Sab enrichment culture exhibited robust N_2O reduction activity, metagenomic investigations of Sab subcultures did not generate evidence for the presence of clade I or clade II C-*nosZ*.



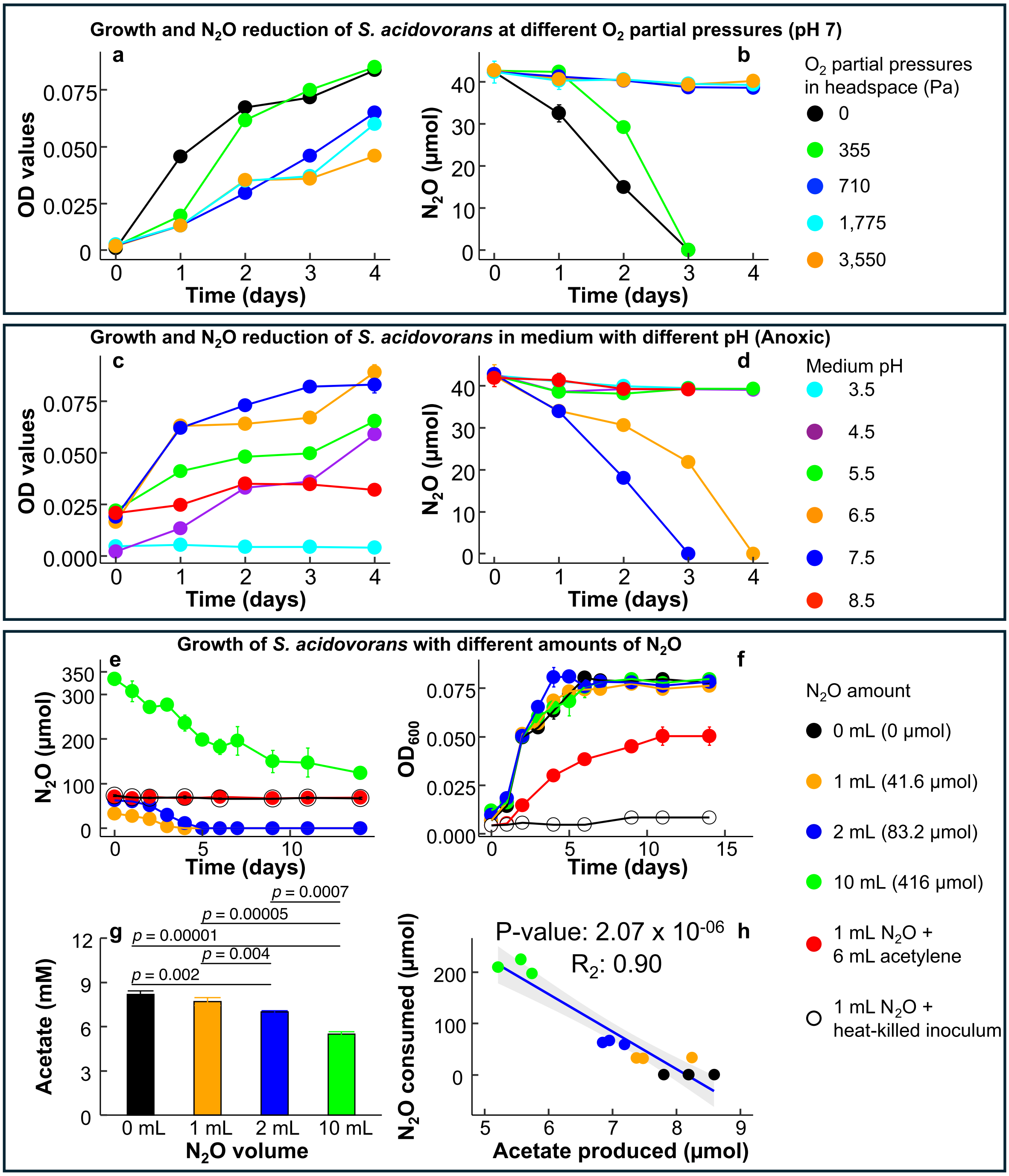
Extended Data Fig. 2 | Fumarate, H_2, and N_2O utilization in Sab enrichment cultures following 25 sequential transfers. Panels **a-e** show fumarate consumption and the formation of acetate, propionate and succinate in medium amended with (**a**) fumarate, (**b**) fumarate + H_2 + N_2O, (**c**) fumarate + H_2 + N_2O + TSB, (**d**) fumarate + H_2 + TSB, and (**e**) fumarate + N_2O + TSB. Panels **f-i** depict H_2 and N_2O consumption in medium that received (**f**) fumarate + H_2 + N_2O, (**g**) fumarate + H_2 + N_2O + TSB, (**h**) fumarate + H_2 + TSB, (**i**) fumarate + N_2O + TSB. Data shown represent averages and standard deviations of triplicate cultures. Error bars are not visible when smaller than the symbols.



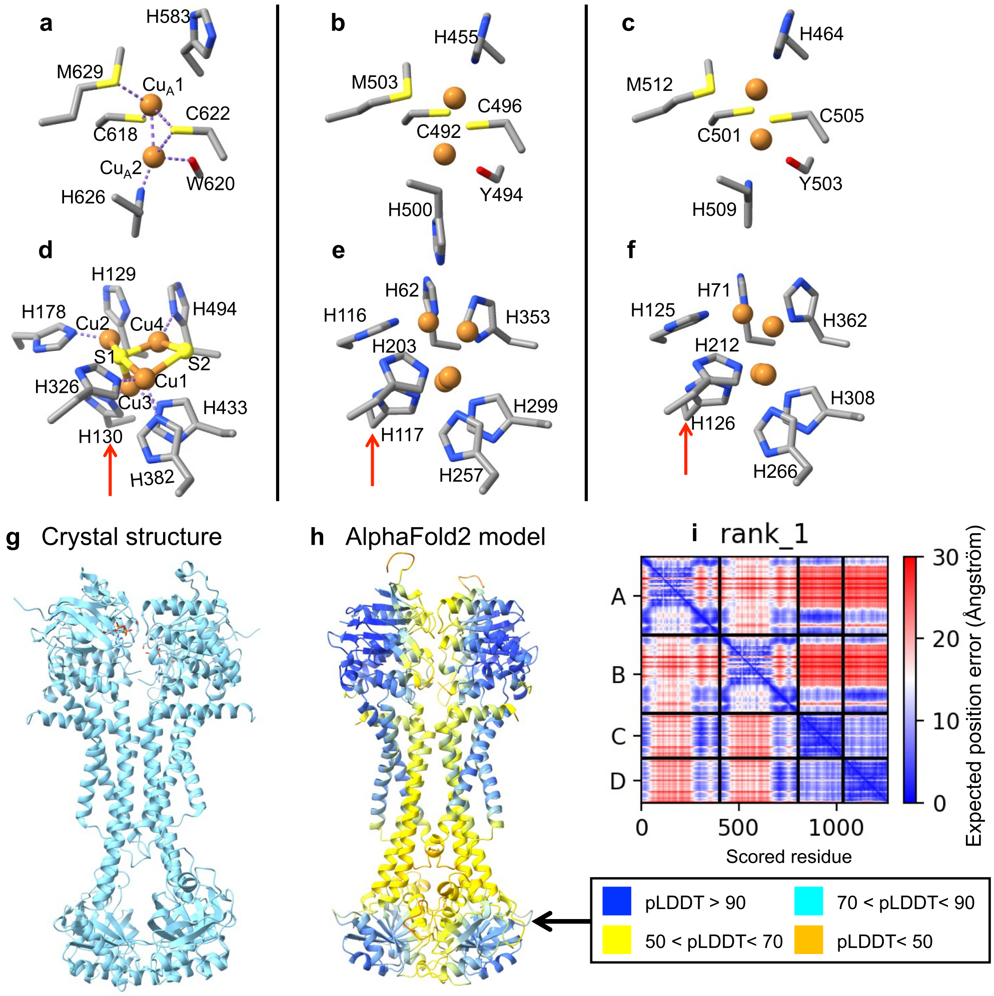
Extended Data Fig. 3 | Phylogenomic analysis revealing the taxonomic affiliation of *Desulfitobacterium nosdiversum* strain Sab5. (**a**) The phylogenomic analysis used 120 conserved bacterial marker genes and included *Desulfitobacteriaceae* genomes available in the NCBI database. The scale bar indicates 0.05 nucleotide substitutions per site. Bootstrap (BP) values are divided into the categories BP < 70 (solid black circles) and BP ≥ 90 (solid grey circles). (**b**) Line graph showing Single Copy Protein- (SCP-) based average amino acid identity (AAI) between the strain Sab5 genome and individual genomes included in the phylogenomic analysis, with the scale displayed on the x-axis. Values for each comparison between the strain Sab5 genome and related genomes are represented by black dots, with the comparison to itself being 100%.



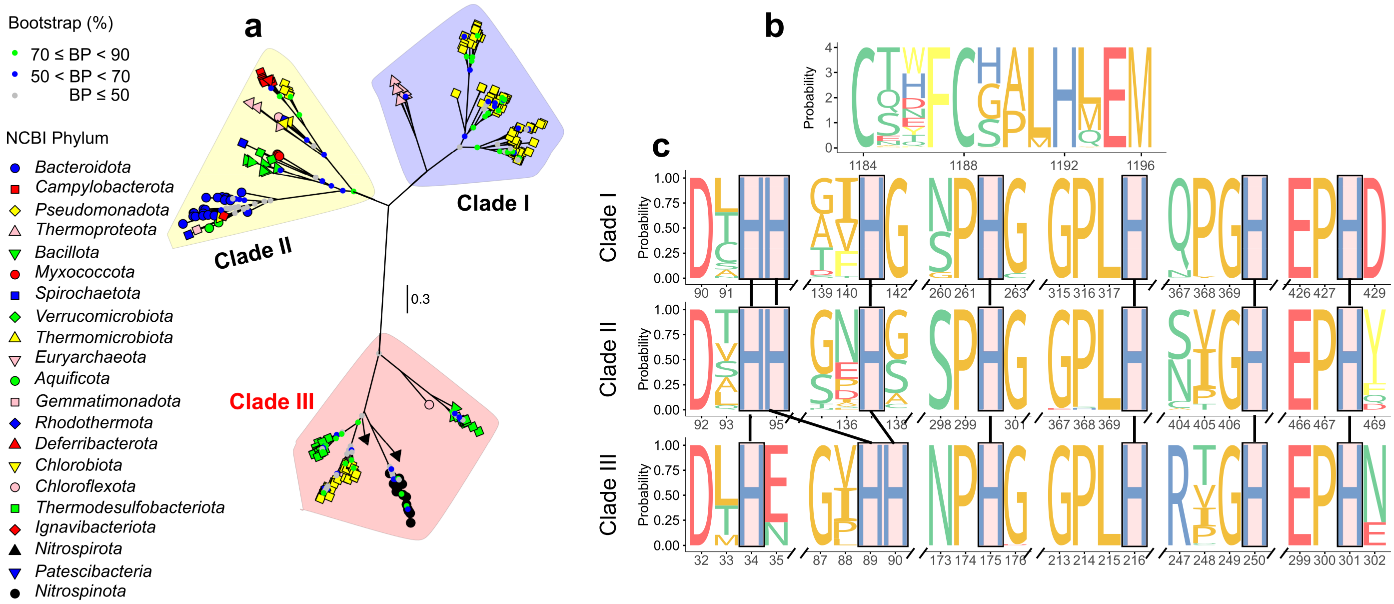
Extended Data Fig. 4 | Growth yield of *Desulfitobacterium nosdiversum* strain Sab5 and protein expression during growth with N_2O. (**a**) Growth yields were estimated by correlating L-*nosZ* gene copy numbers with amounts of N_2O consumed. Cell suspension samples were collected following complete consumption of N_2O from triplicate vessels. Strain Sab5 cell numbers were determined with a qPCR assay targeting the L-*nosZ* gene. The genome of strain Sab5 contains a single L-*nosZ* gene, and L-*nosZ* gene counts equal the cell numbers. Sab enrichment cultures produced $(6.8 \pm 0.77) \times 10^9$ *Desulfitobacterium nosdiversum* strain Sab5 cells mmol N_2O^{-1}, equivalent to a growth yield (dry weight) of 4.21 ± 0.48 mg mmol N_2O^{-1} (assuming a dry weight of 6.13×10^{-10} mg per cell[27]). The gray shading indicates the 95% confidence intervals around the mean values. (**b**) Metaproteomic analysis assigned 1,720 proteins to *Desulfitobacterium nosdiversum* strain Sab5 cells collected from Sab enrichment cultures grown with 5 mM fumarate, 0.02 g L^{-1} TSB, 4.16 mM (nominal) H_2, and 4.16 mM (nominal) N_2O for 3 days when >50% of the initial N_2O dose had been consumed. L-NosZ, MacB type ATP-binding cassette (ABC) transporter, *c*-type cytochrome (K03888), *b*-type cytochrome (K00412), AtoC (K07714) and SasA (COG4191) were all detected (red arrows), in addition to a protein of unknown function encoded by a gene neighboring L-*nosZ*. The curved line shows the Log_2 transformed protein abundances, and the vertical lines represent the standard deviation of triplicate cultures.



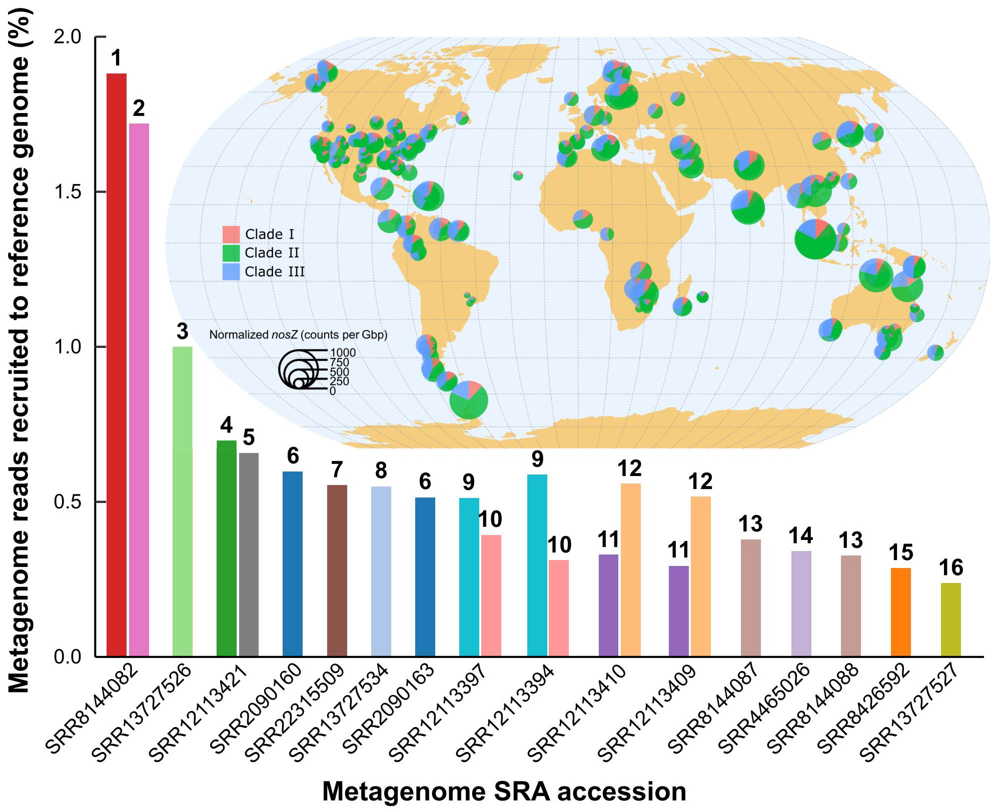
Extended Data Fig. 5 | Growth and N_2O reduction in cultures of *Sporomusa acidovorans* strain Mol. Panels **a-d** show growth of *Sporomusa acidovorans* strain Mol and N_2O consumption at different O_2 partial pressures and pH conditions. (**a**) Growth of strain Mol at pH 7 with aqueous phase O_2 concentrations of 0, 4.55 μM (355 Pa), 9.1 μM (709 Pa), 22.75 μM (1,780 Pa), and 45.5 μM (3,550 Pa). (**b**) N_2O consumption in bottles with different O_2 partial pressures at pH 7. The growth medium was prepared anoxically, and O_2 was introduced by injecting 0, 1, 2, 5, 10 ml of air (i.e., 21% O_2, v/v). (**c**) Growth of strain Mol in medium with the pH adjusted to 3.5, 4.5, 5.5, 6.5, 7.5, and 8.5. (**d**) N_2O consumption observed at the different medium pH values. Panels **e-h** depict strain Mol cultures grown in basal salt medium amended with 1 g l^{-1} yeast extract in the presence of different amounts of N_2O. (**e**) N_2O consumption, (**f**) growth, and (**g**) acetate production in *Sporomusa acidovorans* cultures, with each set of triplicate cultures receiving different amounts of N_2O. Acetate production was measured in samples withdrawn from cultures on day 14. (**h**) Correlation between N_2O consumption and acetate yields. A linear regression was performed (blue line with 95% confidence interval), yielding a significant positive correlation ($R^2 = 0.90$, $p = 2.07 \times 10^{-6}$). No N_2O consumption was observed in control vessels that received autoclaved inocula or in live cultures amended with 6 mL (v/v, 10%) acetylene. Data shown represent averages and standard deviations of triplicate cultures. Error bars are not shown when smaller than the symbols. Pairwise comparisons were performed using two-sided Student's t-tests with *p* values adjusted for multiple testing using the Holm-Bonferroni method[120].



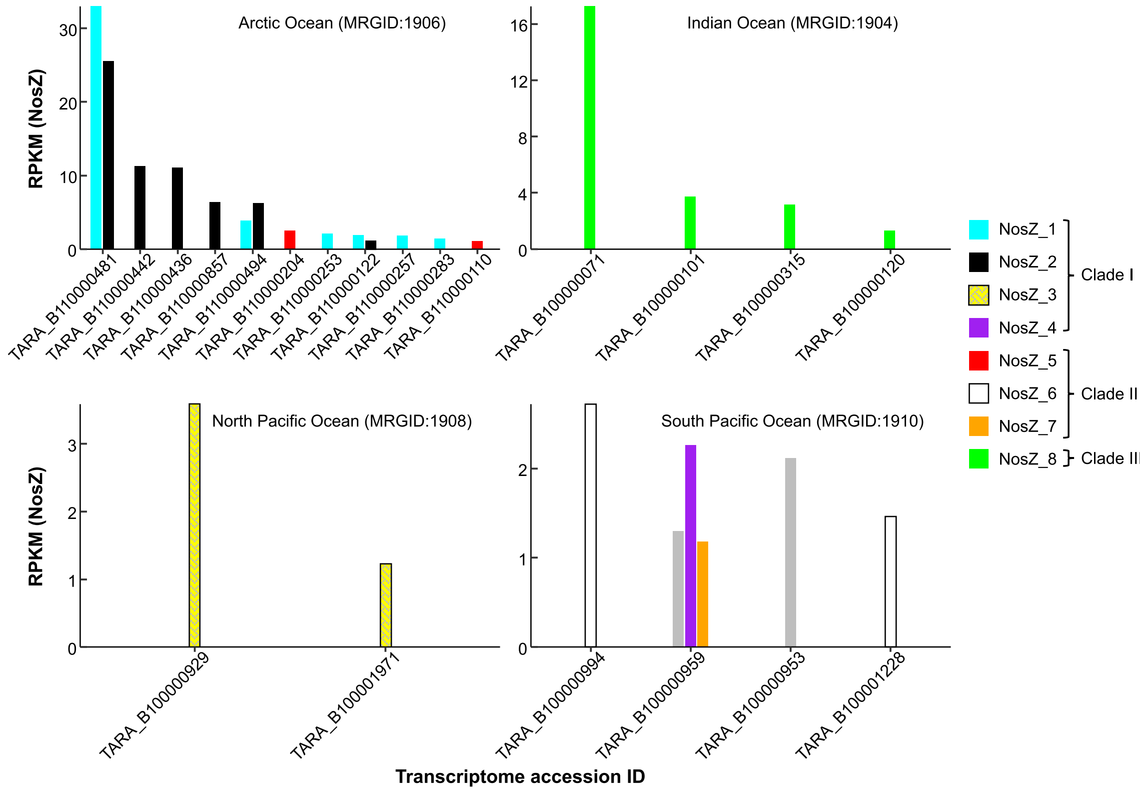
Extended Data Fig. 6 | AlphaFold models of L-NosZ and the MacB-type ABC transporter. Panels **a-f** show Cu-binding active sites in C-NosZ derived from a 1.7 Å X-ray crystal structure of *Pseudomonas stutzeri* (PsNosZ, PDB entry 3SBQ) and representative AlphaFold3 models of apo L-NosZ of *Desulfitobacterium nosdiversum* strain Sab5 and *Sporomusa acidovorans* strain Mol. (**a**) Cu_A site in *Pseudomonas stutzeri* NosZ; (**b**) putative Cu_A site in *Desulfitobacterium nosdiversum* strain Sab5 NosZ; (**c**) putative Cu_A site in *Sporomusa acidovorans* strain Mol NosZ; (**d**) the Cu_Z site in *Pseudomonas stutzeri* NosZ; (**e**) putative Cu_Z site in *Desulfitobacterium nosdiversum* strain Sab5 NosZ; (**f**) putative Cu_Z site in *Sporomusa acidovorans* strain Mol NosZ. The root-mean-square deviations (RMSDs) between paired H130 in 3SBQ and either H117 in *Desulfitobacterium nosdiversum* strain Sab5 L-NosZ or H126 in *Sporomusa acidovorans* strain Mol NosZ (indicated by red arrows) are higher than other paired histidine residues because H117 and H126 originate from different ß-strands in the C-NosZ and L-NosZ enzymes, respectively. Although the position and relative orientation of this histidine residue differ, it is expected to coordinate Cu_3 (see panel **d**) in a similar manner. Specifically, H129 and H130 in the PsNosZ X-ray structure are located on the same ß-strand. In comparison, H116 and H117 in the model of NosZ from *Desulfitobacterium nosdiversum* strain Sab5 are located on the same ß-strand. Similarly, H125 and H126 in *Sporomusa acidovorans* strain Mol NosZ model come from the same ß-strand. Panels **g-h** show a crystal structure and AlphaFold model of MacB-type ABC transporter. (**g**) Homodimeric crystal structure of the ABC transporter MacB from *Acinetobacter baumannii* (PDB entry 5WS4). (**h**) AlphaFold2 model of the putative MacB transporter encoded on the *Desulfitobacterium nosdiversum* strain Sab5 genome generated using ColabFold with 5WS4 as a template. The model is a dimer of homodimers comprising four monomers, i.e., two copies of MacB and two copies of the ABC-type transporter. (**i**) Predicted aligned error (PAE) for the MacB transporter protein chains A, B, C, D, with lower PAE values (blue) indicating higher model confidence. The predicted local distance difference test (pLDDT) scores, a per-residue measure of local confidence, are indicated by colors.



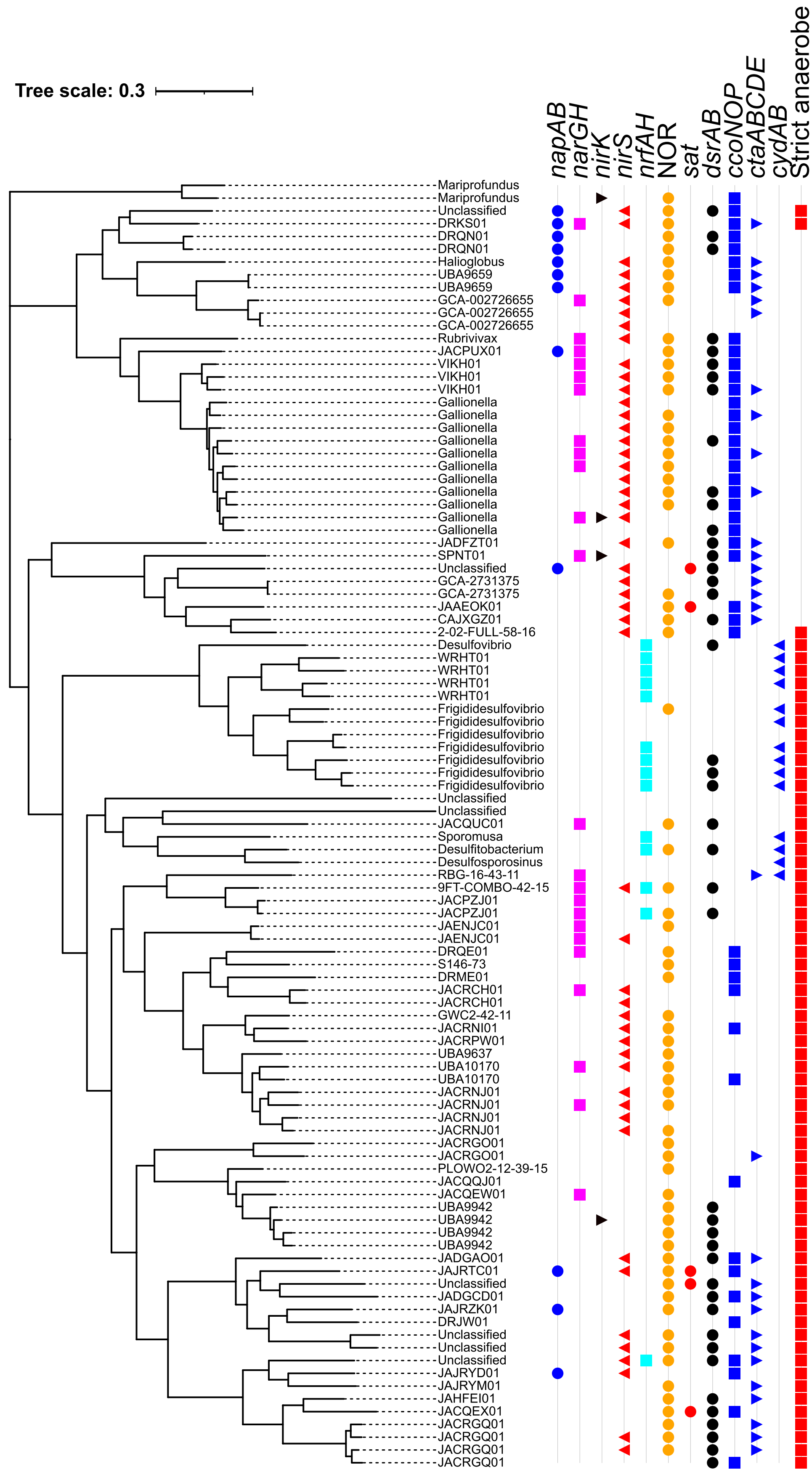
Extended Data Fig. 7 | Features differentiating L-NosZ from C-NosZ sequences. (**a**) Maximum likelihood phylogenetic tree based on protein sequences of clade I C-NosZ (n = 105), clade II C-NosZ (n = 68) and clade III L-NosZ (n = 95). The phylogenetic analysis included 1,451 sites. The colored dots at the nodes indicate the RaxML-ng bootstrap values; bootstrap values > 90% are not shown. The tree scale bar represents the mean number of amino acid substitution per site. (**b**) The Cu_A motifs involved in electron transfer to the Cu_Z site are 100% conserved (CXXFCXXXHXEM) in C-NosZ and L-NosZ. (**c**) Independent alignments of C-NosZ and L-NosZ sequences showing differences in conserved histidine residues associated with the Cu_Z site. The copper binding sites, Cu_A and Cu_Z, were inferred based on analysis of X-ray crystal structure data[121]. The height of the symbols (i.e., the size of letter abbreviations for amino acids) reflects the relative frequency of the amino acid at that position. The multiple sequence alignment plots were created with ggmsa[104].



Extended Data Fig. 8 | Population genome-resolved and metagenomic read-resolved surveys of the environmental distribution of L-*nosZ*. Bars show metagenomic reads (%) recruited to the clade III L-NosZ-encoding genomes. The x-axis displays the metagenome accession IDs. Relevant information on metagenomes and genome bins is summarized in Supplementary Tables 13 and 14, respectively. The numbers above each bar indicate the MAGs used as reference genomes in metagenome recruitment analysis. The 16 MAGs detected along with the highest taxonomic classification are represented by numbers. (1) GCA009782775 (*Desulfovibrionaceae*); (2) GCA015231995 (*Nitrospinota*); (3) GCA016223025 (*Gallionellaceae*); (4) GCA015232395 (*Magnetococcales*); (5) GCA008933335 (*Casimicrobiaceae*); (6) GCA001804905 (*Nitrospinota*); (7) GCA011052055 (*Nitrospinota*); (8) GCA016214995 (*Desulfobacterota*); (9) GCA016200645 (*Desulfobacterota*); (10) GCA024244705 (*Alphaproteobacteria*); (11) GCA011051655 (*Chromatiales*); (12) GCA016218475 (*Gallionella*); (13) GCA031269345 (*Frigididesulfovibrio*); (14) GCA030739775 (*Alphaproteobacteria*); (15) GCA003450795 (*Desulfobacterota*); (16) GCA016191995 (*Rubrivivax*). Individual genome bins are distinguished by colors. The world map inset shows the geographic distribution of 1,053 metagenomes representing 12 distinct biomes from six continents. The pie charts indicate the normalized clade I (red), clade II (green), and clade III (blue) *nosZ* counts relative to the total *nosZ* counts in each metagenome, with the radii indicating the total *nosZ* counts per metagenome. A *nosZ* phylotype was inferred based on the phylogenetic placement on the GraftM NosZ reference tree (Extended Data Fig. 7).



Extended Data Fig. 9 | *nosZ* transcripts detected in the Tara Oceans metatranscriptome (Ocean metaT) dataset[53]. To explore *nosZ* expression, an HMM was built from 173 C-NosZ and 95 L-NosZ sequences to search the protein sequences provided in the Ocean metaT dataset. Transcripts with expression levels higher than 1 RPKM were considered expressed and are represented in the graphs. The search and filtering criteria resulted in a total of four clade I-type sequences (NosZ transcripts 1-4), three clade II-type sequences (NosZ transcripts 5-7), and one clade III-type sequence (NosZ transcript 8). Relevant metadata (e.g., dissolved O_2 concentrations) corresponding to the metatranscriptomes were obtained from the Ocean metaT dataset. The taxonomic assignments of each NosZ transcript are: NosZ_1 (*Gammaproteobacteria*); NosZ_2 (*Gammaproteobacteria*); NosZ_3 (Unclassified bacteria); NosZ_4 (Unclassified bacteria); NosZ_5 (*Flavobacteriaceae*); NosZ_6 (*Bacteroides*); NosZ_7 (Unclassified bacteria); NosZ_8 (*Nitrospinota*). The dissolved O_2 concentrations varied in the Tara Oceans sampling locations: Arctic Ocean (200 μM < c[O_2] < 400 μM); Indian Ocean (c[O_2] < 4 μM); North Pacific Ocean (c[O_2] < 3 μM); South Pacific Ocean (3 μM < c[O_2] < 250 μM). The transcriptome accession IDs correspond to the Ocean metaT dataset.



Extended Data Fig. 10 | See next page for caption.

Extended Data Fig. 10 | Potential utilization of electron acceptors by bacteria represented by genome bins harboring clade III L-*nosZ* and their predicted O_2 preference. The tree was constructed using the GTDB-TK workflow[79]. The scale bar represents the mean number of amino acid substitution per site. Terminal reductases investigated include nitrate reductase (encoded by *napAB* or *narGHIJ*), NO-forming nitrite reductase (*nirK* or *nirS*), ammonium-forming nitrite reductase (*nrfAH*), sulfate adenylytransferase (*sat*), and dissimilatory sulfite reductase (*dsrAB*) based on the annotations derived from eggNOG database v5[122]. Seven curated HMMs, comprising genes encoding nitric oxide reductases (NOR, NO → N_2O)[123], were queried against the protein sequences. The resulting NOR sequences were then filtered by conserved motif sequences in the different NOR families (i.e., cNOR, qNOR, gNOR, nNOR, sNOR, bNOR and eNOR). Genes encoding bacterial high affinity (*ccoNOP* and *cydAB*) and low affinity (*ctaABCDE*) oxidases catalyzing the reduction of O_2 reduction were also investigated[58]. Bacteria represented by genome bins were classified as facultative and strict anaerobes with genome-based computational models using GenomeSpot v1.0.1[59], and the red squares indicate strict anaerobes. About two-thirds (n = 70) of the genome bins harbor NOR genes. Seventeen genome bins contain all genes required for complete denitrification ($NO_3^- \rightarrow NO_2^- \rightarrow NO \rightarrow N_2O \rightarrow N_2$).

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Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Inficon EZ IQ software (version 3.3.2 SP2) for GC data. OpenLab ChemStation for HPLC data. Applied Biosystems ViiA 7 system (Applied Biosystems, Waltham, MA, USA) for qPCR data. AxioVision for light microscope image acquisition. Proteome Discoverer v2.3.0.523 for proteomics data.

Data analysis

FastQC v0.11.9 for read quality assessment. Fastp v0.20.1 for metagenome quality filtration. Megahit2 v1.2.9 for assembly of Illumina sequence data. Maxbin v2.2.7 and Metabat v2.15 for binning. Bin refinement via DAS_Tool v1.1.6. CheckM2 v1.0.2 for analysis of bin quality. (meta)Flye v2.9.5 for assembly of Nanopore (for Sporomusa acidovorans) and PacBio (for Sab enrichment culture) sequence data. Canu (v2.2) for PacBio read correction. Polypolish v0.6.0 for sequence polishing. Circlator v1.5.5 for construction of closed circular contig. GTDB-TK v2.3.2 for taxonomic placement of genome bins. FastAAI for amino acid identity calculation. FastANI v1.34 for nucleotide identity calculation. Sequence alignment via MAFFT. trimAl v1.4.1 for alignment curation. ModelTest-NG v0.1.7 for evolutionary model optimization. Phylogenetic analyses via RAxML and FastTree. FoldTree for building structural phylogenies. ggmsa for visualization of multiple sequence alignment. Automated functional annotation via COG-emapper. Prodigal v2.6.3 for translation of nucleotide sequence to protein sequence. Refinement of annotation via NCBI conserved domain database. SignalP6.0 for prediction of signal peptides. AlphaFold2 and AlphaFold3 for structure prediction. ChimeraX v1.9 for AlphaFold model visualization. Ggtree package v3.15.0 and iTOL v6 for tree visualization. cblaster v1.3.20. itol.toolkit package v1.1.9 for compiling phylogeny associated metadata. clinker v 0.0.31 and gggenomes package for gene cluster analysis. RecruitPlotEasy package for evaluating genome coverage evenness in metagenomic datasets. CoverM v0.7.0 for metagenomic reads recruitment analysis. GraftM v 0.15.0 for detecting metagenomic reads. HMMER v3.4 for homologous protein analysis. pplacer V1.1 for placing metagenomic reads on phylogenetic trees. ggplot2 package for data visualization. ggscatterpie for creating scatterpie plots on geographical map. Bowtie2 v2.4.2 for reads mapping. In house scripts integrating publicly available codes are deposited in https://github.com/utguang/Clade3-NosZ.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The Sab enrichment culture metagenomic datasets (both Illumina and PacBio data) and the resulting closed genome of Desulfitobacterium sp. Sab5 are deposited under NCBI BioProject ID PRJNA1176277. The Sabana microcosm metagenomes are publicly available in NCBI Bioproject ID PRJNA901179. Illumina and PacBio datasets used for construction of a circular genome of Sporomusa acidovorans strain Mol are publicly available under NCBI BioProject ID PRJNA310710. Genome bins analyzed in this study are from Genome Taxonomy Database (https://gtdb.ecogenomic.org), a global soil genome bins database (https://microbma.github.io/project/SMAG.html), and an ocean genome bins database (https://doi.org/10.6084/m9.figshare.c.5564844.v1). The publicly available genomes and raw metagenomes analyzed in this study are publicly available in the NCBI database. Lists of the datasets are summarized in Supplementary Tables 7, 8, 10-14, 16, 17. The mass spectrometry proteomics raw data and sequence files are available at the ProteomeXchange Consortium via the MassIVE database with the unique dataset identifier MSV000096274. The metaproteomic search results are publicly available through the MassIVE database (ftp://massive-ftp.ucsd.edu/v07/MSV000096274/).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Not applicable.
Reporting on race, ethnicity, or other socially relevant groupings	Not applicable.
Population characteristics	Not applicable.
Recruitment	Not applicable.
Ethics oversight	Not applicable.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes were not predetermined. All data presented are laboratory-based investigations, except for the genomic and transcriptomic surveys and the AlphaFold-based structure modeling. Statistics and conclusions are based on data from biological replicates (at least triplicates) with data generally obtained from at least two independent experiments. An exception is the cultivation of axenic Desulfitobacterium spp. cultures, which included biological duplicates to test N2O reduction capacity. Negative controls (i.e., heat-killed cultures) were included. No statistical methods were used to predetermine sample sizes for the genomic and transcriptomic surveys or the AlphaFold-based structural modeling. Sample sizes were determined based on the availability publicly accessible metagenomes, genome bins, and transcriptomic datasets relevant to the environments of interest. The inclusion of thousands of metagenomes and genome bins (as detailed in Supplementary Tables) ensures broad ecological representation for detecting the distribution and expression patterns of clade III nosZ genes. For AlphaFold modeling, representative sequences were selected to span the phylogenetic and structural diversity of NosZ. This selection enabled confident modeling of structural features without the need for large-scale sampling. The chosen sequences are representative of major clades observed in the genomic survey and align with the functional validation work.
Data exclusions	No data were excluded from the analyses.
Replication	Cultivation-based experiments were performed in triplicate (or more replicates) and confirmed in at least three independent experiments, unless otherwise stated. All attempts at replication were successful and yielded consistent results. An exception is the incubation of axenic Desulfitobacterium spp. cultures, which were performed in duplicate cultures; these results were also reproducible. Microscopy-based observations were reproduced across three independent preparations. Quantitative assays (e.g., metabolite measurements, gene abundance) were conducted with biological replicates (n = 3 per condition). Detailed replicate numbers are provided in the figure legends or relevant Methods subsections. All attempts at replications were successful.
Randomization	Randomization was not relevant for this study. Each experiment included controls and treatments.
Blinding	Blinding was not relevant to this study, which was driven by explorative enrichment culturing aimed at cultivating and characterizing microorganisms with nitrous oxide-reducing capabilities. All experimental conditions—including medium composition, electron donor/acceptor combinations, and incubation atmospheres—were deliberately designed and implemented by the researchers to probe specific physiological traits. Outcomes such as microbial growth, substrate consumption, and N_2O reduction were assessed using objective, quantitative measurements (e.g., gas chromatography, HPLC, qPCR), eliminating the potential for observer bias. Each experiment was conducted with clearly defined treatment groups and included appropriate negative and, when available, positive controls to ensure interpretability and reproducibility.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A